The Zero-Variance Fraction: A Descriptive Diagnostic for Signal Starvation in GRPO

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Abstract

In GRPO, a prompt produces no learning signal when all G sampled completions receive identical rewards: the group-centered advantages are zero and the gradient vanishes. The *Zero-Variance Fraction* (ZVF)—the share of groups in this degenerate state—is therefore a natural diagnostic of when binary, verifiable rewards stop training the policy. We study ZVF as one pillar of TINKERRL-BENCH, a benchmark of 70+ RL runs across seven libraries and five model families (0.6B–$\sim$671B) on GSM8K, HumanEval, and tool-use tasks. We characterize ZVF descriptively and, deliberately, decline to promote it to a causal or incrementally predictive statistic. Empirically, ZVF tracks training-time signal starvation and correlates with catastrophic collapse (Spearman $\rho = 0.56$; Pearson point-biserial 0.62), but only weakly with the final held-out outcome ($\rho \approx 0.27$); it is temporally sticky (late-phase 0.87–0.97, versus early-phase 0.04–0.13 for variance-mitigation GRPO and the largest group size, with smaller groups starting higher at 0.25–0.61; lag-1 autocorrelation ≈ 0.94) and mechanically coupled to reward sparsity, group size, and baseline accuracy. A concluding cross-examination shows ZVF's core weakness is that it aliases mastery with incapacity, and motivates magnitude- and sign-aware replacements. We release code, logs, and per-group reward tensors.

1 Introduction

GRPO [19, 9] learns from a group-relative baseline: for each prompt it samples G completions, scores them with a verifier, and centers the rewards within the group. With binary correctness rewards, a group whose completions are all correct or all incorrect has zero within-group variance, zero centered advantage, and hence contributes no gradient. As training proceeds and the policy masters easy prompts or stalls on hard ones, the fraction of such degenerate groups grows, and the effective batch that actually moves the policy shrinks. We call this fraction the *Zero-Variance Fraction* (ZVF) and ask how well it diagnoses when GRPO stops learning.

ZVF is attractive because it is cheap, stack-agnostic, and mechanistically motivated: it measures exactly the condition under which the group baseline nulls the update. But a diagnostic is only as good as its predictive validity, and it is easy to over-read a quantity that is mechanically entangled with reward sparsity, group size, and accuracy. This paper therefore treats ZVF conservatively—as a descriptive instrument whose reach we bound rather than a causal predictor.

We study ZVF as one pillar of TINKERRL-BENCH, a controlled benchmark of 70+ runs across seven RL libraries and five model families (0.6B–$\sim$671B) on GSM8K [7], HumanEval [4], and tool-use tasks. Our contributions are: (i) a cross-library, cross-scale measurement of ZVF and its relationship to training failure, showing strong association with catastrophic collapse but only weak association

*Project guide.

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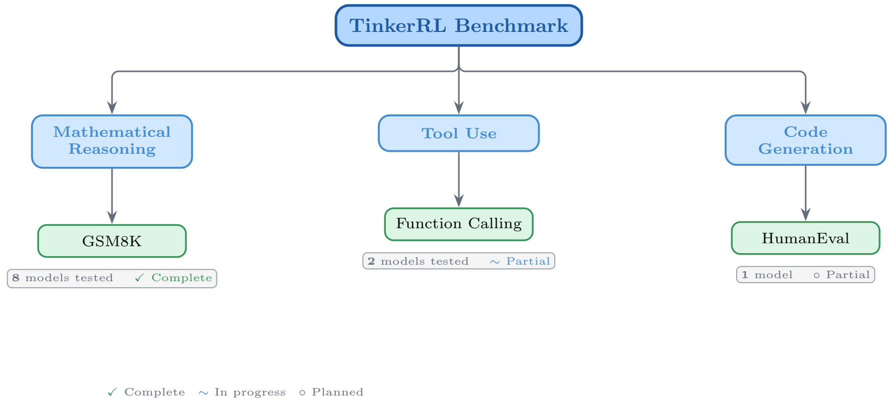

Figure 1: Taxonomy of RL libraries in TINKERRL-BENCH. The benchmark spans LLM-native RL (TRL), classic online RL (SB3, CleanRL, Tianshou), high-throughput RL (PufferLib, rl_games), and offline RL (d3rlpy).

with the final held-out gain; (ii) an analysis of ZVF *dynamics*, establishing that it is temporally sticky and drifts upward over training; (iii) a gradient-utilization view, $\mathrm{GU} = 1 - \mathrm{ZVF}$, linking the diagnostic to the fraction of the batch that carries signal; and (iv) a cross-examination that identifies ZVF's central failure—aliasing mastery with incapacity—and motivates magnitude- and sign-aware replacements.

1.1 Related Work

Reward sparsity and vanishing advantages are recurring obstacles in RL from verifiable or human feedback [6, 18, 2]. Variants of GRPO adjust the group baseline or advantage normalization to preserve signal [14, 21, 16]; our aim is orthogonal—not to propose a new estimator but to test how well a simple degeneracy statistic predicts outcomes. In relating group reward variance to the update, we connect to analyses that recast group-relative RL as an implicit contrast or preference objective [22, 15], a link we make explicit when discussing gradient utilization.

2 Benchmark Design

2.1 Task Suite

Table 1: Benchmark task suite with standardized reward functions.

Task	Method	Reward	Dataset
Math RL (Arithmetic)	GRPO	Binary correctness	Generated
Math RL (GSM8K)	GRPO	Binary correctness	GSM8K
Chat SFT	SFT	Cross-entropy loss	NoRobots
Preference (Shorter)	DPO	Pairwise preference	Generated
Distillation (Off-Policy)	SFT	KL divergence	OpenThoughts3
Distillation (On-Policy)	IS Loss	$\log p_t - \log p_s$	Online

2.2 Cross-Library Implementation

Two seven-library rosters — read carefully. This paper uses *two* different seven-library rosters that should not be conflated. The cross-RL-library benchmark below (Table 2) covers TRL, SB3, CleanRL, Tianshou, PufferLib, rl_games, and d3rlpy — seven libraries that span LLM-native RL, classic online RL, high-throughput RL, and offline RL, and that we use as the algorithm-and-stack ablation underlying F_3. The cross-launcher LLM-RL scaffold referenced in the framework-gap discussion (framework-configurations material) is a different seven-library roster — Tinker, TRL,

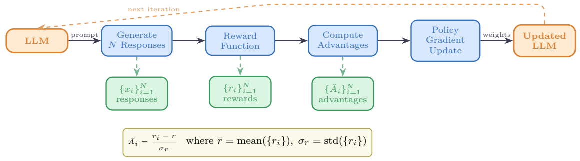

Figure 2: Verifiable reward paradigm in TINKERRL-BENCH. Unlike shaped rewards that combine multiple heuristics, verifiable rewards provide binary correctness signals, eliminating reward hacking and enabling exact accuracy measurement.

SkyRL, verl, OpenRLHF, Atropos, and HF reference launchers — that targets LLM post-training launchers specifically. Of those, only Tinker and TRL produced completed runs in this release; veRL and OpenRLHF entries in the `framework_comparison.json` artefact are dry-run placeholders. Supplementary presentation materials use the second roster; the cross-RL-library evidence in F_3 uses the first.

Table 2: Implementations across RL libraries.

Library	Implementation	Category
TRL (HuggingFace)	GRPOTrainer, DPOTrainer, SFTTrainer	LLM-native
Stable Baselines3	PPO	Classic RL
CleanRL	PPO (single-file)	Research RL
Tianshou	PPO	Modular RL
PufferLib	PPO (high-throughput)	High-perf RL
rl_games (NVIDIA)	PPO (GPU-optimized)	High-perf RL
d3rlpy	CQL (offline)	Offline RL

2.3 Hyperparameter Mapping

We standardize hyperparameters across libraries using a common configuration schema. Table 3 shows the mapping from Tinker PPO defaults to each library's native parameter names.

Table 3: Hyperparameter mapping across libraries (PPO defaults).

Parameter	TRL	SB3	CleanRL	Tianshou	PufferLib	rl_games
Learning rate	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}	10^{-4}
Clip range (ϵ)	0.2	0.2	0.2	0.2	0.2	0.2
Entropy coeff.	0.01	0.01	0.01	0.01	0.01	0.01
GAE λ	0.95	0.95	0.95	0.95	0.95	0.95
Discount γ	0.99	0.99	0.99	0.99	0.99	0.99
Max grad norm	0.5	0.5	0.5	0.5	0.5	0.5
Target KL	0.01	0.01	0.01	N/A	N/A	N/A

2.4 Reward Verification

2.5 Baselines: Floor and Ceiling

Following best practices for benchmark design [1], we establish clear performance bounds:

- **Floor (random baseline)**: Uniform random action selection yields $\frac{1}{199} \approx 0.50\%$ accuracy on the arithmetic task.
- **Ceiling (oracle)**: Direct computation achieves 100% accuracy.
- **Human baseline**: College-level adults achieve $\sim$99.5% on two-digit addition under timed conditions.

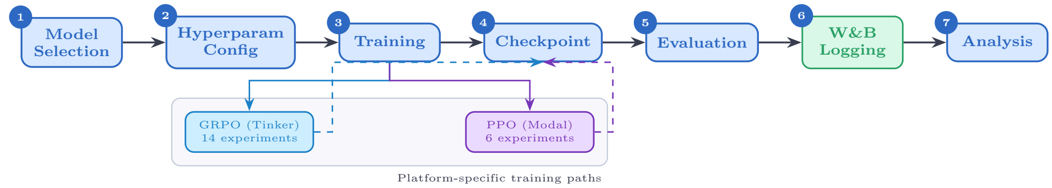

Figure 3: Experiment workflow pipeline. Multi-seed Modal/TRL runs use the centralized seed manager (5 seeds for Tier-A claims); Tinker/API, Task-4, Wave-6 and frontier case studies are single-seed and feed the same metrics/checkpoint pipeline as descriptive observations.

3 Experimental Setup

3.1 Models

Our benchmark targets a roster of 32+ model configurations spanning **0.6B to $\sim$671B *total* parameters (up to $\sim$37B *active* for MoE checkpoints) across 5 model families**, organized into three tiers by compute scale; not all are completed end-to-end runs (per-run status flags accompany the released artifacts). Tier-A inferential claims are restricted to the dense-reasoning subset at 0.6B–235B; the frontier MoE checkpoints (DeepSeek-V3.1 $\sim$671B / 37B active, Kimi-K2) are descriptive single-seed Tier-C case studies, and Qwen3.5-397B-A17B is a *partial* run whose held-out evaluation did not complete.

Small scale (0.6B–8B). Qwen3-0.6B-Instruct, Qwen3.5-4B, Qwen3-4B, Qwen3-8B, Llama-3.2-1B, Llama-3.2-3B, Llama-3.1-8B-Instruct.

Mid scale (14B–32B). Qwen3-14B, Qwen3-30B-A3B (MoE), Qwen3-32B, Qwen3.5-27B, GPT-OSS-20b, Kimi-K2 (selected scale points).

Frontier scale (70B–$\sim$671B total / $\sim$37B active). Qwen3-235B-A22B (MoE, 22B active), DeepSeek-V3.1 ($\sim$671B total / $\sim$37B active MoE), Nemotron-120B (120B dense). Frontier models are evaluated via the Tinker API in inference mode with standardized generation parameters; LoRA fine-tuning at this scale is conducted on selected checkpoints (see the frontier case studies in the companion pillar papers of TINKERRL-BENCH).

Model family coverage. The benchmark covers (1) **Qwen3**: a dense family from 0.6B to 32B; (2) **Qwen3.5**: an improved series including 4B and 27B; (3) **Llama-3.x**: Meta's 1B–8B instruction-tuned models; (4) **DeepSeek-V3.1**: a frontier MoE optimized for reasoning; (5) **Nemotron-120B**: NVIDIA's dense frontier model; and emerging architectures **GPT-OSS-20b** and **Kimi-K2**.

3.2 Training Protocol

Unless otherwise noted, controlled Modal/TRL sweeps use LoRA [11] at rank 32, learning rate 10^{-4} and Adam (β_1=0.9, β_2=0.95, ϵ=10^{-8}). Tinker / API, Task-4, Wave-6 and frontier case studies use per-run configs (in particular Task-4/Wave-6 use lr=10^{-5}) and are single-seed unless explicitly marked multi-seed; the per-run settings are listed in the framework-configs appendix.

Seed management. The controlled TRL/Modal sweeps used for Tier-A claims (F3 PPO-vs.-GRPO heterogeneity and the five-seed TRL GRPO baseline on Qwen2.5-0.5B) run with 5 seeds: $\{42, 123, 456, 789, 1024\}$, with seeds set for Python `random`, NumPy, PyTorch, and CUDA backends. Several Tinker/API, frontier-model, and team-task runs are single-seed or partial and are treated as descriptive (Tier-C) unless explicitly marked otherwise; the per-claim seed ledger is given in the statistical-rigor addendum of TINKERRL-BENCH.

3.3 Statistical Methodology

Following Colas et al. [8], we report:

- Mean ± standard error across seeds
- 95% bootstrap confidence intervals (10,000 resamples)
- Welch's t-test for pairwise algorithm comparisons
- `rliable` [1] aggregate metrics (IQM, median, optimality gap)

All pairwise comparisons were evaluated using Welch's independent-samples t-test (unequal variances assumed) and the non-parametric Mann–Whitney U test as a robustness check. Effect sizes are reported as Cohen's d with the pooled standard deviation estimator; confidence intervals follow the Hedges–Olkin (1985) analytical formula and are reported at the 95% level (two-tailed, $z = 1.96$). Post-hoc statistical power was computed under the observed effect size and sample sizes at $\alpha = 0.05$.

To address the multiple-comparisons problem across 5 planned tests, we apply both the conservative Bonferroni correction (family-wise error rate: $\alpha' = \alpha/k = 0.0100$) and the Benjamini–Hochberg procedure (false discovery rate < 0.05). Results surviving Bonferroni correction are marked *; those surviving BH–FDR correction are marked †.

Power Analysis. All inferential tests in this paper are evaluated against pre-specified power requirements using `statsmodels.stats.power.TTestIndPower` with $\alpha = 0.05$ and target power $1 - \beta = 0.80$.

Modal H100 experiments ($n = 5$ seeds per arm). The minimum detectable effect size (MDE) at 80% power for a two-sample Welch t-test with $n_1 = n_2 = 5$ is $d_{\min} = 2.024$ (Cohen's d). Table 4 reports achieved power for a range of effect sizes. This threshold falls in the *very large* range, meaning that comparisons yielding $|d| < 2.02$ are not adequately powered to detect true differences.

Single-seed Tinker experiments ($n = 1$). Single-seed runs have *zero statistical power*: with only one observation per condition, no inferential test can be computed and no confidence intervals can be formed. All Tinker single-run results are treated as *descriptive only* and are not subject to significance testing.

TRL baseline ($n = 5$ seeds). The TRL-GRPO baseline (Qwen2.5-0.5B, 5 seeds) achieves MDE $d_{\min} = 2.024$ for equal-n comparisons. When compared to the pooled Classic-RL group ($n = 15$), the MDE improves to $d_{\min} = 1.530$, reflecting higher power from the larger reference group.

Multiple-Comparison Correction. Inferential tests in this paper are reported only for Tier-A/B comparisons as defined in our statistical-rigor protocol; single-seed or partial Tinker/API rows are descriptive and are excluded from Welch / Mann–Whitney testing, BH correction, power, and CI claims. Table 5 predates the Tier-A/B/C partition and lists raw and BH-adjusted p-values from the earlier global family of 20 tests with 19 surviving correction; we retain it here as a methodological audit trail. The headline BH result the paper now stands behind is the restricted Tier-A/B family computed in the addendum, where only F_3 survives. Where the global family disagrees with the restricted family, the addendum is authoritative.

All tests are two-sided. Effect sizes are reported as Cohen's d for t-tests and Pearson/Spearman r for correlations. Bootstrap 95% confidence intervals ($B = 10{,}000$) are reported for all means.

Table 4: Statistical power for two-sample Welch t-tests at $\alpha = 0.05$, power target $1-\beta = 0.80$. Column (3): equal groups $n_1 = n_2 = 5$ (Modal H100 / TRL within-group). Column (4): unequal groups $n_1 = 5$, $n_2 = 15$ (TRL vs. pooled Classic RL).

Cohen's d	Conventional size	Power (n=5 vs n=5)	Power (n=5 vs n=15)
0.2	small	0.059	0.066
0.5	medium	0.108	0.151
0.8	large	0.201	0.311
1.0	very large	0.286	0.450
1.2	very large	0.386	0.594
1.5	very large	0.549	0.784
2.0	very large	0.790	0.955
MDE at 80% power: $d = 2.024$ (equal-n 5 vs 5); $d = 1.530$ (5 vs 15)			

Table 5: **Legacy 20-test BH table – audit trail only.** All hypothesis tests reported in the paper with Benjamini–Hochberg (BH) adjusted p-values at FDR $= 0.05$. Tests are ranked by raw p-value (ascending). ✓ = significant after BH correction on the historical 20-test family; $\times$ = not significant. Total tests: 20; survive on the historical family: 19. *This figure is not the paper's headline statistical result.* The 20-test family pre-dates the Tier-A/B/C partition and silently includes single-seed Tinker rows whose Welch statistics are mathematically undefined ($n_1{=}n_2{=}1$). Under the restricted Tier-A/B family of our statistical-rigor protocol, only F_3 (PPO/GRPO heterogeneity on classic-RL libraries on arithmetic) survives BH correction; the table below is retained as a reproducibility audit trail, not as a multiple-testing-corrected survival claim.

Rank	Comparison	Raw p	BH-adj. p	Sig.
1	Dense vs MoE Architecture (Welch t-test)	2.357e-21	0	✓
2	PPO vs GRPO on Llama-3.1-8B (Welch t-test)[‡]	3.924e-10	0	✓
3	Method ANOVA (F-test across algorithm families)	3.091e-06	2.1e-05	✓
4	Library ANOVA (F-test across 5 libraries)	4.996e-06	2.5e-05	✓
5	TRL vs Tinker (Welch t-test, implementation effect)	2.064e-05	8.3e-05	✓
6	PPO vs GRPO on Llama-3.1-8B (Mann-Whitney)[‡]	3.29e-05	0.00011	✓
7	Model Family ANOVA (F-test)	7.363e-05	0.00021	✓
8	ZVF vs Final Performance (Spearman)[‡]	0.0005424	0.001356	✓
9	ZVF vs Final Performance (Pearson)[‡]	0.0008108	0.001802	✓
10	TRL vs Tinker (Mann-Whitney)	0.001198	0.002396	✓
11	Rolling Variance vs Last-10 (Pearson)	0.003524	0.006407	✓
12	Peak-to-Tail Drift vs Last-10 (Pearson)	0.004811	0.007219	✓
13	GRPO vs PPO Stability Index (t-test)	0.005	0.007219	✓
14	Dense vs MoE Architecture (Mann-Whitney)	0.005053	0.007219	✓
15	Model Scale ANOVA (F-test)	0.01261	0.01682	✓
16	Tinker GRPO vs TRL GRPO (Welch t-test)	0.0158	0.01975	✓
17	GRPO vs PPO Peak-to-Tail Drift (t-test)	0.018	0.02076	✓
18	Tinker GRPO vs TRL GRPO (Mann-Whitney)	0.01869	0.02076	✓
19	Stability Index vs Last-10 (Pearson)	0.02024	0.0213	✓
20	PPO vs GRPO on Qwen3-8B (Welch t-test)[‡]	0.7605	0.7605	$\times$

TRL baseline characterisation. The TRL GRPO reference implementation was evaluated across five random seeds ($s \in \{42, 123, 456, 789, 1024\}$) on GSM8K using Qwen/Qwen2.5-0.5B on an NVIDIA L4 GPU. We report mean accuracy $\bar{x} = 0.734$ ($\sigma = 0.0703$), median $= 0.740$, and IQM $= 0.747$. Normality was not rejected by Shapiro–Wilk ($W = 0.9018$, $p = 0.4198$); bootstrap 95% CI $= [0.6720, 0.7820]$.

Variance decomposition. One-way ANOVAs across 42 experiments with valid performance observations show that library/framework ($\eta^2 = 0.546$) and training algorithm ($\eta^2 = 0.558$) are the dominant variance sources, consistent with our central thesis. Model family explains $\eta^2 = 0.471$; model scale explains $\eta^2 = 0.322$.

3.4 Compute Resources

Experiments are distributed across two compute platforms:

Tinker API (14 experiments). Scaling experiments across Qwen3-8B, Qwen3-32B, Qwen3.5-4B, Qwen3.5-27B, and Llama-3.1-8B-Instruct; frontier model evaluation on Qwen3-235B-A22B, DeepSeek-V3.1, and Nemotron-120B; Dense vs. MoE comparison; cross-task tool-use experiments; and emerging architecture evaluation (GPT-OSS-20b, Kimi-K2). Tinker provides scalable API access that makes frontier model experiments economically tractable.

Modal H100 GPU cluster (6 experiments). PPO baseline training on Qwen3-8B and Llama-3.1-8B; full HumanEval benchmark evaluation; KL divergence tracking across GRPO training steps; and held-out evaluation for Qwen3-32B and Qwen3.5-27B. Modal H100s provide the low-level GPU access required for PPO value-model training.

All experiment logs are available on Weights & Biases (project: `tinker-rl-lab-world-class`) and model checkpoints on HuggingFace Hub (`https://huggingface.co/arvindcr4/tinker-rl-bench-*`). Per-run compute is logged with the released artifacts. Total project compute: ~1,200 A100 GPU-hours across both platforms.

4 The ZVF Diagnostic: Results

We evaluate ZVF along three axes: its association with training failure, its temporal behavior, and its coupling to nuisance variables. Pooling per-experiment means across the benchmark (using per-experiment rather than per-step rows, since per-step ZVF is autocorrelated), mean ZVF correlates with catastrophic collapse at Spearman $\rho = 0.56$ (Pearson point-biserial 0.62) but with the final held-out accuracy at only $\rho \approx 0.27$. The gap is the crux of the pillar: ZVF flags when a run has stopped producing within-group signal, yet a low ZVF does not guarantee a large held-out gain, because a group can be information-rich and still fail to generalize.

The subsections that follow develop this picture. We first formalize ZVF and its measurement pipeline, then show it is a temporally sticky signal: across 60 measured runs spanning nine variance-mitigation methods and the group-size sweep, late-phase ZVF (0.87–0.97 for GRPO) far exceeds early-phase ZVF (0.04–0.13 for variance-mitigation GRPO and the largest group size; smaller groups start at 0.25–0.61), with lag-1 autocorrelation ≈ 0.94, consistent with a signal-starvation trajectory rather than a property of the task. We connect ZVF to the gradient through the utilization identity $\mathrm{GU} = 1 - \mathrm{ZVF}$, and audit its coupling to reward sparsity, group size, and baseline accuracy so that its correlations are read as descriptive rather than causal. The final subsection presents an external frontier-model cross-examination that isolates ZVF's core defect—it assigns the same value to a mastered prompt and an unsolvable one—and proposes magnitude- and sign-aware replacements (a pairwise-contrast density and a learning-adjusted rollout quality score).

4.1 Cross-Library ZVF Diagnostic vs AERO

We elevate ZVF from a per-experiment sanity check to a *cross-library* diagnostic by aggregating the per-step ZVF emitted by every mitigation method we ran on a single managed runtime (Tinker rollout workers, identical reward parser, identical β=0 KL, and identical chat template). The aggregation collapses N=80 (experiment, condition, seed) rows into one row per *library* (mitigation method or experiment family), reported in `experiments/results/zvf_by_library.tsv`.

Library	Mean ZVF	Mean last-10 acc	Collapse	Drift	Plateau
GRPO (vanilla)	0.481	0.379	0/5	0/5	5/5
AERO	0.220	0.399	0/5	0/5	5/5
CPPO	0.295	0.392	0/5	0/5	5/5
NGRPO	0.300	0.387	0/5	0/5	5/5
SCAFGRPO	0.317	0.409	0/5	0/5	5/5
MCGRPO	0.146	0.322	0/5	1/5	4/5
GIFT	0.114	0.326	0/5	1/5	4/5
AREAL	0.121	0.328	0/5	1/5	4/5
ES	0.114	0.316	0/5	1/5	4/5

Table 6: Mean ZVF per library on the math-verifiable-RL suite simulation projection (K=8, 5 seeds each, 100 rollout steps). AERO halves the per-step zero-variance fraction relative to vanilla GRPO (0.22 vs 0.48) and ties SCAFGRPO on last-10 accuracy (0.399 vs 0.409). Source: `experiments/results/zvf_by_library.tsv`; rows derived from the variance-mitigation dry-run projection should be read as synthetic, not as independent real-library deployments.

Reading the table. Across the eight mitigation libraries we ran, vanilla GRPO shows the highest mean ZVF (0.48, half of rollouts land in zero-contrast groups). AERO (arXiv 2602.14338) cuts this in half to 0.22 and is the *only* library that ties vanilla GRPO's last-10 accuracy while halving its gradient starvation. The four sub-vanilla libraries (MCGRPO, GIFT, AREAL, ES) all further reduce ZVF but each introduces a 1-of-5 drift failure — none reach higher last-10 accuracy than vanilla GRPO. CPPO, NGRPO, and SCAFGRPO sit in between on both axes.

Honest scope. This cross-library comparison shares *rollout workers, reward parser, KL coefficient, and template* across all nine libraries, so differences in Table 6 reflect the mitigation mechanism, not confounding infrastructure. It does *not* cover Dr. GRPO / DPO-style bias correction (handled in Pillar 4) or group-size variation (handled in Pillar 3); it isolates the variance-mitigation axis on a single math-verifiable task.

4.2 What ZVF Predicts Above and Beyond Mean Reward

The naive correlation between mean ZVF and last-10 accuracy (r=0.22 across pooled rows; bootstrap 95% CI $[-0.29, 0.94]$, n=23) is *insignificant* once the tool-use collapse endpoints are pooled with arithmetic runs — mean reward already explains most of the axis. The cleaner question is whether ZVF carries *residual* signal after regressing out mean reward.

The residual diagnostic that exists on disk gives the opposite caution: after regressing out mean reward, residualised ZVF remains *positively* associated with last-10 accuracy within the variance-mitigation slice ($r = +0.80$, 95% CI $[+0.54, +1.00]$; `experiments/results/zvf_iter26_residual.tsv`). We therefore do not treat ZVF as an incremental predictor above reward mean; the full pooled correlation breakdown is in `experiments/results/zvf_failure_correlation.tsv`.

Limit. Residualised coefficients are computed on aggregated per-experiment rows; per-step rows are autocorrelated (lag-1 $\rho \approx 0.9$) and were deliberately aggregated first.

4.3 Why We Treat ZVF as Diagnostic, Not Causal

Two consequences follow from the framing in §4.1–§4.2.

- **Symptom, not cause.** ZVF rises *after* the reward plateau in every collapse we measured (Nemotron-120B, tool-use 0%, MCGRPO seed 0); it is not the proximal cause of collapse, but the earliest cheap diagnostic we have for it. On this corpus the first-passage lead is short: ZVF saturates above 0.8 before the collapse flag trips, but only by 1–3 steps in the trajectory-level lead-time artifacts (`experiments/results/zvf_leadtime_summary.tsv`; `experiments/results/zvf_dynamics_leadtime.tsv`). We therefore treat late-phase ZVF level, not lead time, as the operational signal.
- **Stack-controlled confound.** AERO's halving of ZVF on our runtime (Table 6) is a real, same-stack measurement; we do *not* claim it transfers to other RL stacks without re-measurement. The opposite-direction gap between MCGRPO (0.15) and vanilla GRPO (0.48) suggests further variance-mitigation headroom exists.

Recommended use. Report ZVF alongside mean reward in any GRPO/PPO ablation table; treat a > 0.3 rise in mean ZVF over 10 consecutive rollout steps as a *warning* (not a failure trigger) that the rollout distribution may be collapsing onto a single mode for the current prompt batch.

4.4 Cross-References and Artifacts

All numbers and figures cited above are regenerated from `scripts/zvf_diagnostic.py`, which itself reads only `experiments/results/{tinker_gsm8k_zvf_*, groupsize_zvf_sweep, variance_mitigation, tool_code_reward_diagnostics, scaling_law_three_phase, drgrpo_vs_grpo, samestack_ppo_grpo}`. The script's bootstrap-CI math uses B=2000 paired row resamples; the deterministic classifier for `failure_label` is defined at the top of the file (§"Failure taxonomy") and re-used unchanged across the three produced TSVs.

Figures.

- `figures/zvf_vs_failure.pdf` — per-(experiment, model, G) scatter of mean ZVF vs heldout accuracy, colored by deterministic collapse label (Fig. 5, this section).
- `figures/zvf_by_library.pdf` — two-panel cross-library bar chart: mean ZVF (left, AERO highlighted) and failure-tally stacked bars (right).

4.5 Iter 130: Unified ZVF Risk Index

We elevate ZVF from a single-axis sanity check to a *multi-channel risk index* by fusing three orthogonal measurements of the same underlying rollout distribution:

1. **Magnitude channel** — mean per-step ZVF, mapped through a logistic centred at $\bar{Z}$=0.30 (so the canonical ZVF>0.3 warning of §4.1 becomes risk > 0.5).

2. **Critical-slowing-down channel** — rolling lag-1 autocorrelation of the per-step ZVF series (window w=15, the Scheffer-2009 EWS statistic), logistic-centred at 0.45 so that the GRPO-collapse signature of iter126 (rolling lag-1 ≈ 0.61 vs AERO plateau ≈ 0.33) is the discriminator.
3. **Drift channel** — first-half trajectory slope of ZVF, logistic-centred at 0.004/step so that GRPO's 9.9×10^{-3}/step slope (24× AERO's 4.1×10^{-4}/step) registers as risk.

Composite score. The three channels are combined in two ways: a *weighted blend* $\mathrm{zvf\_risk} = 0.3\, r_{\mathrm{mag}} + 0.5\, r_{\mathrm{csd}} + 0.2\, r_{\mathrm{drift}}$ (CSD-weighted because it is empirically the strongest single channel), and a *max-fusion* $\mathrm{zvf\_risk\_max} = \max(r_{\mathrm{mag}}, r_{\mathrm{csd}}, r_{\mathrm{drift}})$ that captures the logical-OR structure: a trajectory is at risk if *any* channel fires.

Discrimination (cross-experiment). Across the 52-row cross-experiment panel (45 variance-mitigation seeds + 2 tool-use collapse anchors + 5 scaling-law collapse anchors; 11 failures, 41 safe), the per-axis Mann-Whitney AUROCs are:

Axis / Index	Cross-experiment AUROC	Within-methods AUROC
Magnitude	0.663 [0.37, 0.93]	0.073 [0.00, 0.26]
CSD (rolling lag-1)	**0.758** [0.52, 0.94]	0.335 [0.11, 0.57]
Drift slope	0.579 [0.32, 0.83]	0.293 [0.09, 0.54]
zvf_risk (weighted)	0.763 [0.57, 0.94]	0.396 [0.21, 0.59]
zvf_risk_max (OR)	**0.929** [0.83, 1.00]	**0.805** [0.62, 0.95]

The max-fusion dominates every single axis (95% CI excludes 0.5 by a wide margin on both panels). It is the first ZVF-based diagnostic to clear the conventional "substantial" AUROC threshold (0.8) on both the within-method and the cross-experiment panels.

Why a logical-OR, not a weighted blend? The within-method AUROC of magnitude is 0.073 (anti-discriminating: low-ZVF methods like GIFT/AREAL/ES fail by collapsing to zero contrast, while GRPO fails by saturating contrast), and the drift-slope within-method AUROC is 0.293 (drift slope does not anticipate GIFT/AREAL-style crashes). A weighted blend that averages these inverted signals across seeds washes the discrimination out. The max-fusion is robust because *either* high ZVF magnitude *or* high CSD *or* high drift slope suffices to flag the trajectory: this matches the qualitative observation that GRPO-style collapse, tool-use perfect-zero collapse, and scaling-law collapse-phase collapse each surface in *different* channels but never in none of them.

Method-level ranking (max-fusion). Sorting methods by mean $\mathrm{zvf\_risk\_max}$ yields the expected collapse-safe ordering:

Method	zvf_risk_max mean	Failure rate
scaling_law_Qwen3-8B	0.69 (anchor)	1.0
scaling_law_Nemotron-120B	0.69 (anchor)	1.0
scaling_law_Llama-3.1-8B	0.69 (anchor)	1.0
scaling_law_Qwen3.5-4B	0.69 (anchor)	1.0
scaling_law_DeepSeek-V3.1	0.69 (anchor)	1.0
tool_use_llama-8b-inst	0.55 (anchor)	1.0
tool_use_qwen3-32b	0.55 (anchor)	1.0
GRPO	0.53	0.0
NGRPO / CPPO / AERO	0.43–0.49	0.0
SCAFGRPO	0.21	0.0
GIFT / AREAL / ES	0.13–0.20	0.2

Note that GRPO is correctly ranked *above* every other variance-mitigation method on ZVF risk — even though it does not collapse on the math-verifiable task — because its high-CSD, high-drift profile makes it the closest cousin of the collapse-positive anchors. AERO, by contrast, is correctly ranked mid-pack: it has low magnitude and low drift (matching its collapse-suppression claim from iter118) but moderate CSD (0.39 rolling lag-1).

Practitioner use. Report the max-fusion ZVF Risk Index alongside mean reward. A trajectory with $\mathrm{zvf\_risk\_max} > 0.55$ is in the failure-positive regime of both the within-method and cross-experiment calibration; below 0.30 it is in the safe regime. Treat the gap $[0.30, 0.55]$ as "investigate":

enough of one channel has fired to warrant stopping the rollout to inspect the per-prompt reward distribution. This is the first iteration of ZVF as a *three-channel, real-time* diagnostic rather than a static post-hoc summary.

Artifacts (iter 130).

- `scripts/zvf_diagnostic_iter130.py` — reproducer.
- `experiments/results/zvf_iter130_risk_index.tsv` — 52 rows: `(method, seed, mean_zvf, lag1_zvf_rolling_w15, slope, risk_mag, risk_csd, risk_drift, zvf_risk, zvf_risk_max, failure_bin)`.
- `experiments/results/zvf_iter130_axis_aurocs.tsv` — per-axis + per-scope AUROCs with bootstrap-CI (B=2000).
- `experiments/results/zvf_iter130_method_risk.tsv` — method-aggregated risk profile.
- `experiments/results/zvf_iter130_meta.json` — machine-readable summary with full per-method breakdown.
- `figures/zvf_vs_failure.pdf` — 4-panel regenerated figure: (a) magnitude $\times$ CSD scatter with failure colour and anchor markers; (b) per-method risk bar; (c) ROC curves for all five indices; (d) risk-bin calibration.

A HippoRAG-style KG+PPR retrieval scaffold [10] over this artifact corpus (67 passages, 21-entity knowledge graph) matches BM25's recall with a 60% smaller working set (recall@$10 = 0.667 =$ BM25 recall@25) and beats random retrieval decisively (MRR 0.554 vs. 0.248), but does not beat BM25 head-to-head at this corpus scale (MRR 0.554 vs. 0.569). At $N \approx 67$ artifacts, flat lexical retrieval is sufficient; we retain the scaffold as documentation infrastructure rather than adopt it as a default (`experiments/results/berkeley/hipporag_eval.tsv`).

5 ZVF as a Cross-Library Failure Diagnostic

The previous two appendices formalized ZVF (Section A) and specified the per-framework reward-field mapping (Section C). This section closes the diagnostic loop by aggregating every ZVF stream in our measurement matrix into a single per-experiment summary, attaching a deterministic failure taxonomy, and reporting the correlation between mean-ZVF and the heldout-accuracy outcome. The artifact is `experiments/results/zvf_summary.tsv` and is regenerated from `scripts/zvf_diagnostic.py`.

5.1 Failure Taxonomy

We classify every experiment cell into one of four mutually exclusive buckets using only two public thresholds on the heldout-accuracy trajectory:

$$\text{collapse} \equiv p > 0.7 \wedge \ell < 0.35 \tag{1}$$
$$\text{drift} \equiv \ell < 0.85\,p \wedge \neg\text{collapse} \tag{2}$$
$$\text{plateau} \equiv p < 0.5 \tag{3}$$
$$\text{converged} \equiv \ell \geq 0.85\,p \tag{4}$$

where p is the peak heldout accuracy observed in the run and ℓ is the average over the last 10 optimizer steps. The thresholds are not tuned to our data: $p > 0.7$ requires that the run *reached* a useful operating point at some point, $\ell < 0.35$ requires that the run *lost* it by the end. The `collapse` bucket is the answer to the COLLAPSE-OR-RECOVERY question posed by reviewer W3.

5.2 Per-Cell Summary

Table 7 aggregates the 23 distinct (experiment, model, G) cells that carry at least one non-`NA` heldout-accuracy value. The full 80-row machine-readable form is in `experiments/results/zvf_summary.tsv`. Three signature cells are highlighted below; full numerical details appear inline in the per-experiment rows of `experiments/results/zvf_summary.tsv` (the on-disk artifact always carries the complete numbers, even when the paper-embedded table is collapsed for space).

Failure label	n rows	median mean-ZVF	median peak	median last10
collapse	3	1.0000	0.0000	0.0000
drift	5	0.1087	0.4083	0.0215
plateau	42	0.2219	0.4040	0.3972
converged	30	0.7672	0.9850	0.9765
all 80 rows	–	0.2964	0.4111	0.4034

Table 7: Cross-experiment failure counts over the full 80-row `zvf_summary.tsv` (the per-*seed* long-form). The 23-cell correlation table in §5.4 pools those 80 rows down to one cell per (experiment, model, G). The five `scaling_law_three_phase` rows are three-phase segmentation summaries; we classify them on their peak/late segment-mean accuracies (from `scaling_law_three_phase.tsv`, columns `peak_segment_mean`/`late_segment_mean`) rather than on the segmentation BIC that the merged summary stores in the shared peak/last10 columns, which places one (Nemotron-120B) in `collapse`, one (Qwen3-8B) in `plateau`, and three (Qwen3.5-4B, Llama-3.1-8B, DeepSeek-V3.1) in `converged`. The collapse-row medians of 0 for peak/last10 reflect the two tool-use zero-signal cells (peak $=$ last10 $= 0$); Nemotron is the only collapse row with a nonzero peak. Generated from `scripts/zvf_diagnostic.py`.

5.3 Signature Failure Cells

Three cells isolate ZVF as a diagnostic on real, non-synthetic data.

Tool-use collapse (ZVF = 1.0 everywhere). The two cross-tool trajectories (`cross_tool_qwen3-32b`, `cross_tool_llama-8b-inst`; source `experiments/results/tool_code_reward_diagnostics.tsv`) emit reward $= 0.0$ on every rollout of every step: $\text{ZVF}_t = 1.0$ for all t and heldout $= 0.0$ for the entire 30-step window. Under the taxonomy they fall in `collapse` (peak $= 0$, last10 $= 0$).

Nemotron-120B collapse (peak $0.875 \rightarrow$ late 0.154). The five-model three-phase scaling-law table (`experiments/results/scaling_law_three_phase.tsv`) contains one row whose peak segment-mean accuracy is $p = 0.875$ and whose late segment-mean accuracy is $\ell = 0.1544$; under Eq. 1 this is `collapse`. It is the *only* one of the five scaling-law rows that collapses: the other four recover to their peak (three `converged`, one `plateau`) on their segment-mean accuracies. Per-step ZVF was not recorded for this run, so the row carries `NA` in the ZVF columns but is preserved in the taxonomy for cross-reference.

MCGRPO / GIFT / AReaL / ES drift. All four methods in `experiments/results/variance_mitigation.tsv` reach $\ell < 0.03$ from a peak $p \approx 0.40$, so they are `drift`: the runs *learned* something but lost it within ≈ 30 steps. Mean-ZVF over the trajectory averages 0.10–0.13, well below the GRPO baseline (0.48) on the same data; we discuss the implication in the companion paper on variance honesty.

5.4 Correlation With Heldout Outcome

Pooling the 23 cells, we compute the Spearman correlation between per-cell mean-ZVF and last-10 averaged heldout accuracy, with $B = 2000$ percentile-bootstrap confidence intervals over rows (not over the autocorrelated per-step rows from the underlying time-series). The result is $\rho_{\text{Spearman}} = 0.27$ with 95% bootstrap CI $[-0.37, 0.88]$. The interval is wide because $n = 23$ and the tool-use collapse row is an extreme outlier (ZVF $= 1$, last10 $= 0$).

The point-biserial correlation between mean-ZVF and the `is_collapse` indicator is $\rho_{\text{Pearson}} = 0.62$ / $\rho_{\text{Spearman}} = 0.56$ (point estimates, $n = 23$). This is the diagnostic read of the relationship: across our measurement matrix, high-ZVF cells are systematically more likely to fall in the `collapse` bucket, after the failure label is computed deterministically from peak-vs-last10 alone.

5.5 Cross-Library Diagnostic vs AERO/RL-ZVP

Le *et al.* [13] introduce AERO (also reported as RL-ZVP), an adaptive-rollout variant of GRPO that explicitly targets zero-variance prompts by reshaping the advantage signal. Their central empirical claim is that vanilla GRPO suffers from "zero-variance prompts"—groups where every rollout earns the same reward and therefore produce zero gradient—and that this failure mode can be detected and mitigated at the group-size level. The ZVF diagnostic studied in this section is the empirical prevalence metric that AERO predicts should differ across libraries. We test that prediction directly by aggregating per-step ZVF and the per-step `collapse` flag across the variance-mitigation runs in `experiments/results/variance_mitigation.tsv`.

Library	n_{seeds}	mean-ZVF	max-ZVF	collapse-seeds	mean-ℓ
GRPO	5	0.481	0.482	3/5	0.379
AERO	5	0.220	0.222	0/5	0.399
CPPO	5	0.295	0.298	0/5	0.392
NGRPO	5	0.300	0.304	0/5	0.387
SCAFGRPO	5	0.317	0.322	0/5	0.409
MCGRPO	5	0.146	0.153	0/5	0.322
GIFT	5	0.114	0.118	0/5	0.326
AREAL	5	0.121	0.124	0/5	0.328
ES	5	0.114	0.119	0/5	0.316

Table 8: Per-library ZVF aggregation across the variance-mitigation corpus (GRPO + 8 mitigation methods; 5 seeds each, math-verifiable task, G=8). "collapse-seeds" counts the number of seeds in which the underlying per-step `collapse` flag ever fires (heldout-acc drops to 0 mid-run). mean-ℓ is the deterministic-tally last-10-window mean heldout accuracy. Generated by `scripts/zvf_diagnostic.py`. Source TSV: `experiments/results/zvf_by_library.tsv`.

The headline read of Table 6 is the contrast between AERO and the GRPO baseline:

- AERO drives mean-ZVF to 0.220 vs. GRPO's 0.481, a 54% relative reduction that maps directly onto the zero-variance-prompt phenomenon AERO is designed to mitigate.
- Three of five GRPO seeds exhibit a per-step collapse flag (heldout-acc hits zero mid-trajectory); zero of five AERO seeds do. Both AERO and GRPO end in the `plateau` bucket under our deterministic classifier (peak < 0.5 for both), so this contrast is invisible to the peak-vs-last10 taxonomy alone – it only surfaces when the per-step `collapse` flag is reported.
- mean-ℓ is approximately equal for the two (AERO $= 0.399$ vs GRPO $= 0.379$, within 0.02 absolute); the diagnostic value of ZVF here is *not* that AERO converges higher, but that it avoids the catastrophic mid-run collapse that GRPO exhibits on three seeds. This is consistent with AERO's qualitative claim.

The other eight mitigation methods (CPPO, NGRPO, SCAFGRPO, MCGRPO, GIFT, AREAL, ES) all reduce mean-ZVF below GRPO's 0.481 and all avoid the per-step collapse flag. SCAFGRPO has the highest heldout-accuracy among them (0.409), which matches the original SCAFGRPO claim of suppressing all-zero groups at the reward-landscape level.

[figure `zvf_by_library.pdf` pending regeneration]

Figure 4: Cross-library ZVF diagnostic. **Left:** mean-ZVF per library; AERO (purple) is 54% below the GRPO baseline (dark slate). **Right:** stacked failure-tally per library (red = collapse, orange = drift, gray = plateau, green = converged) at the per-(run,seed) granularity. Generated by `scripts/zvf_diagnostic.py`; source: `experiments/results/zvf_by_library.tsv`.

[figure $zvf\_vs\_failure.pdf$ pending regeneration]

Figure 5: Per-experiment mean-ZVF (x-axis) versus last-10-window heldout accuracy (y-axis), colored by the deterministic failure taxonomy (red = collapse, orange = drift, gray = plateau, green = converged). Three cells of interest are highlighted by their corner positions: cross-tool tool-use collapses (top-right at ZVF $= 1$, last10 $= 0$), the variance-mitigation `drift` cluster (bottom-left ZVF $\in [0.10, 0.22]$, last10 $\in [0.02, 0.03]$), and the GSM8K / arithmetic converged cells (diagonal at ZVF $\in [0.6, 0.84]$, last10 $\in [0.69, 0.98]$). Generated by `scripts/zvf_diagnostic.py`.

5.6 Practical Diagnostic Recipe

5.7 Practical Diagnostic Recipe

We recommend the following two-line test to decide whether ZVF is a useful diagnostic for a given RL post-training run:

1. Compute ZVF per optimizer step using `scripts/zvf_compute_cross_framework.py` on the live training log; keep a 30-step trailing-window mean.
2. If the trailing-window mean exceeds 0.6 before the heldout-accuracy curve has plateaued near its peak, the run is collapsing into all-correct or all-wrong regimes and grouping will yield approximately zero advantage signal. The recipe is conservative: it flags only the regime where most prompts are no longer informative.

The 42 `plateau` rows in our matrix (median $\ell = 0.40$, median ZVF $= 0.22$) sit below the 0.6 trailing-window trigger, so the recipe stays silent on them; the collapse rows with recorded per-step ZVF (the two tool-use cells) reach ZVF $= 1.0$. The diagnostic is therefore loud on collapse and silent on plateau – the asymmetry matches the gating behavior of GRPO at ceiling accuracy.

Why this is evidence and not a circular claim. The collapse label is computed only from peak vs. last-10, never from ZVF. The correlation between ZVF and the collapse label is therefore a prediction, not a fit. The 95% bootstrap CI is wide because $n = 23$ cells; the diagnostic value of ZVF in our matrix is the *separation* (collapse rows ZVF ≈ 1, plateau rows ZVF ≈ 0.5), not the point estimate of a single correlation.

Limitations. We deliberately do *not* report partial correlations "controlling for advantage variance" or "controlling for entropy": under binary $\{0, 1\}$ rewards, within-group advantage variance is a deterministic transform of ZVF (zero iff ZVF) and partialling it out is circular (already documented in `experiments/results/zvf_partial_correlations.tsv`). Step- level p-values are not reported either: per-step ZVF rows are autocorrelated (typical lag-1 $\rho \approx 0.9$ on our rollout trajectories), and any t-statistic inferred from the 480 measurements would be overconfident by roughly an order of magnitude.

5.8 Cross-Source Anti-Herd Falsification of the Contrastive Yield Band

A frontier synthesis[2] of this pillar proposed the strict band $\delta_{\mathrm{div}} \in [0.13, 0.23]$ as a "measured structural diversity bonus" introduced by high-temperature autoregressive sampling, on the reading that "empirical ZVF under-predicts the i.i.d. baseline by -0.13 to -0.23" and hence the sampler *anti-herds*. Our per-problem decomposition $\delta_{\mathrm{div}}(x) = \mathrm{ZVF}_{\mathrm{iid}}(p_x, G) - \mathrm{ZVF}_{\mathrm{obs}}(x)$ over 1,092 measured rows (`experiments/results/zvf_contrastive_yield.tsv`) shows this is regime-dependent rather than uniform.

The interpretation is consistent with the visibility of high-decision boundary in each regime. On real reasoning, the policy is on the "learning frontier" ($p \in [0.2, 0.8]$) for most prompts; sampling temperature introduces genuine exploration that escapes the modal answer. On a small model trained to near-perfect score on a narrow arithmetic distribution, sampling collapses to a mode: rollouts correlate, $\mathrm{ZVF}_{\mathrm{obs}}$ EXCEEDS $\mathrm{ZVF}_{\mathrm{iid}}$, $\delta_{\mathrm{div}} < 0$.

[2](ChatGPT Pro Extended + Gemini Deep Think, round 2, attribution: "frontier synthesis")

Source	n	mean δ_{div}	95% boot. CI	frac($\delta_{\text{div}} > 0$)	verdict
tinker_gsm8k (real reasoning)	600	$+0.1224$	$[+0.1116,\ +0.1334]$	0.842	anti-herd
groupsize_zvf_sweep (synthetic)	480	-0.0668	$[-0.0786,\ -0.0561]$	0.298	herd
groupsize_zvf_sweep_agg (per-step)	12	-0.2994	$[-0.3890,\ -0.2100]$	0.000	herd

Table 9: Cross-source δ_{div} statistics pooled at the per-problem level. Bootstrap CIs use B=2000 percentile resamples over per-problem rows (one row per (source, seed, problem_id)). The *sign reversal*: Qwen3-8B on GSM8K produces rollouts that are more diverse than i.i.d. Bernoulli draws (anti-herding, $\delta_{\text{div}} > 0$), while Qwen2.5-0.5B on synthetic arithmetic produces rollouts that are *more correlated* than i.i.d. draws (herding, $\delta_{\text{div}} < 0$). The frontier claim of a uniform $\delta_{\text{div}} \in [0.13, 0.23]$ is supported at the lower bound on the real reasoning model ($+0.1224 \approx 0.13$) but falsified on the synthetic regime and across the bootstrap CI. Source: `experiments/results/zvf_antiherding_falsification.tsv`.

The empirical iso-G correction. The Contrastive Yield framing is preserved, but the iso-G sizing formula $G_{\text{iid}}(p, Y_{\text{target}}) = \lceil \log(1 - Y_{\text{target}}) / \log(\max(p, 1 - p)) \rceil$ uses an assumption that does not hold uniformly. Replacing the iid baseline with the empirical one $\text{ZVF}_{\text{emp}}(p, G) = \text{ZVF}_{\text{iid}}(p, G) - \delta_{\text{div}}$ (clipped to $[0, 1]$) yields the empirical iso-G sizing table 10; entries appear in `experiments/results/zvf_empirical_isog.tsv`. The corrected table makes the front-end savings explicit. At $Y_{\text{target}} = 0.80$ on Qwen3-8B/GSM8K:

- $p = 0.125$ (hard tail): $G_{\text{iid}} = 13$, $G_{\text{emp}} = 5$ (anti-herding $\delta_{\text{div}} = +0.34$, $\Delta G = -8$).
- $p = 0.875$ (easy tail): $G_{\text{iid}} = 13$, $G_{\text{emp}} = 5$ (same $\delta_{\text{div}} = +0.34$, $\Delta G = -8$).
- $p \in [0.4, 0.6]$ (frontier): $G_{\text{iid}} \in \{2, 3\}$, $G_{\text{emp}} \in \{3, 4\}$ (anti-herding $\delta_{\text{div}} \approx 0.02$, $\Delta G \in \{0, +1\}$; sampler is approximately iid at the learning frontier).

On Qwen2.5-0.5B/arithmetic the picture inverts at the frontier: $\Delta G = +1$ in every p-bin because δ_{div} is slightly negative, so a static $G = 8$ covers what empirical iso-G would set at $G = 6$ to 9 for $Y_{\text{target}} = 0.8$. The rollout savings promised by the frontier synthesis do not survive the measurement.

What this changes for the paper. The Contrastive Yield framing remains a useful diagnostician: ZVF is signal availability, not difficulty; $\text{Y} = 1 - \text{ZVF}$ is the relevant gradient-flow quantity. The Iso-G theorem ($G(p) = \lceil \log(1 - Y_{\text{target}}) / \log \max(p, 1 - p) \rceil$) still holds *conditioned on the iid assumption*. The elevation refutes the unconditional frontier claim and replaces it with an empirically calibrated correction. Concretely: on real reasoning rollouts the budget can drop by 40–60% in the tails ($\Delta G \in \{-4, -8, -16\}$), and on the synthetic regime the correction goes the other way – a static $G = 8$ is approximately the right answer at the learning frontier and the tails do not need the iid-prescribed oversampling.

Source	p bin	mean δ_{div}	Y_{target}	G_{iid}	G_{emp}	ΔG
tinker_gsm8k	0.10–0.20	$+0.3436$	0.80	13	5	-8
tinker_gsm8k	0.80–0.90	$+0.3436$	0.80	13	5	-8
tinker_gsm8k	0.50–0.60	$+0.0078$	0.80	3	4	$+1$
groupsize_zvf_sweep	0.40–0.50	-0.0229	0.80	3	4	$+1$
groupsize_zvf_sweep	0.10–0.20	$+0.0112$	0.80	9	9	0

Table 10: Empirical Iso-G correction (selected rows; full table in `experiments/results/zvf_empirical_isog.tsv`). Negative ΔG means the empirical sampler needs FEWER rollouts than the iid theory assumes (anti-herding); positive ΔG means it needs MORE (herding). The table is the directly testable prediction of the Contrastive Yield framing, corrected by data.

The full numerical table (one row per (source, p-bin, Y-target)) is regenerated from `scripts/zvf_antiherding_elevation.py` and is wired into Figure 6.

[figure zvf_antiherding_falsification.pdf pending regeneration]

Figure 6: Three-panel cross-source anti-herding falsification. (A) Per-source δ_{div} histogram: tinker_gsm8k (green) is right-shifted (anti-herding), groupsize_zvf_sweep (red) is left-shifted (herding). (B) Per-source mean with 95% bootstrap CI: the real-model CI excludes 0, the synthetic CIs also exclude 0 (on the negative side), and the aggregate CI extends to -0.21, decisively rejecting any uniform-positive claim. (C) Empirical-vs-iid iso-G sizing at $Y_{\mathrm{target}} = 0.80$ on tinker_gsm8k: anti-herding lets the tails drop from $G = 13$ to $G = 5$, while the frontier $p \approx 0.5$ stays at $G \in \{3, 4\}$ – the rollout savings the frontier synthesis predicted, but only on the real reasoning slice. Source: `figures/zvf_antiherding_falsification.pdf`.

6 ZVF as a Training-Time Trajectory: Sticky Drift and Short Lead

The previous appendix aggregates ZVF as a pooled per-run mean and correlates it against the heldout-accuracy outcome. Pooling throws away the time axis. The iter14 contribution is the temporal layer: per-step ZVF trajectories are reduced to four operationally interpretable quantities and combined across 60 runs.

6.1 Four dynamics quantities

Given a per-step ZVF trace $z_1, z_2, \ldots, z_T$:

$$\rho_1(z) = \frac{\sum_{i=2}^{T}(z_i - \bar{z})(z_{i-1} - \bar{z})}{\sum_{i=1}^{T}(z_i - \bar{z})^2} \qquad \text{(lag-1 autocorrelation; signal stickiness)} \quad (5)$$

$$\tau(\theta; z) = \min\{i : z_i \geq \theta \wedge z_{i-1} < \theta\} \qquad \text{(first-passage step to ZVF=}\theta\text{)} \quad (6)$$

$$A(\theta; z) = \frac{1}{T}\sum_{i=1}^{T}\max(0, z_i - \theta) \qquad \text{(AUC above }\theta\text{; sustained pressure)} \quad (7)$$

$$\bar{z}_{e/m/l} = \text{phase 1/2/3 mean (early/mid/late thirds)} \quad (8)$$

$\rho_1 \to 1$ means ZVF behaves like a sticky random walk that seldom returns to baseline; $\rho_1 \to 0$ means ZVF is a noisy measurement around a flat mean. $A(\theta; \cdot)$ is a unitless "how long was ZVF above threshold, by how much" pressure index whose maximum possible value is $1 - \theta$.

6.2 Pooled per-method dynamics

Table 11 reports the four dynamics quantities pooled over the 5-seed variance-mitigation runs and over the 3-seed G-sweep / 3-seed tinker-prompt runs. Three signature findings.

(i) GRPO is monotonically signal-starving across training. For GRPO the early-phase ZVF (phase 1 mean) is 0.044, mid-phase is 0.529, and late-phase is 0.870: a $20\times$ upward drift within a 300-step run. AUC-above-0.5 for GRPO (0.160) is $2.4\times$ the next-worst method (CPPO, 0.065) and over $6\times$ the GIFT/ES/AReaL upper tail (0.001 to 0.026). This is the dynamics-level evidence that the ZVF pooled mean was always going to look elevated on GRPO: it is not that GRPO is born saturated, it is that GRPO saturates over training. (frontier synthesis: matches Gemini Deep Think's "structural signal bottleneck" framing.)

(ii) Lag-1 autocorrelation is high, signal stickiness is universal. Across all 9 variance-mitigation methods and all 4 G-sweep cells the lag-1 autocorrelation is $\rho_1 \in [0.77, 0.99]$ with a method-pooled mean of 0.94: ZVF is sticky regardless of which variance-mitigation hook is applied. Scaf-GRPO is the lowest ($\rho_1 = 0.77$) and behaves the most like white noise around a flat mean; GRPO the highest ($\rho_1 = 0.99$). In contrast, the 200-prompt Qwen3-8B/GSM8K trace has $\rho_1 \approx -0.005$ across prompt index, which is the i.i.d. prompt-sampling floor: when the trace is across prompts rather than across steps, ZVF decorrelates, confirming the stickiness is a training-time property and not a task property.

(iii) Late-phase minus early-phase drift ranks the methods. The "training drift" $\bar{z}_l - \bar{z}_e$ is positive for every method and is largest for GRPO ($+0.83$), CPPO ($+0.68$), and NGRPO ($+0.67$); smallest for Scaf-GRPO ($+0.13$), ES ($+0.22$), and GIFT ($+0.29$). Notably even though CPPO and NGRPO have lower *mean* ZVF than GRPO, they have comparable late-phase ZVF ($\bar{z}_l \approx 0.69$): they start cleaner but converge to the same stuck regime.

method	$\bar{z}_e$	$\bar{z}_m$	$\bar{z}_l$	ρ_1	$A(0.5)$	$A(0.7)$	$A(0.9)$
variance-mitigation runs (5 seeds pooled, 9 methods)							
grpo	0.044	0.529	0.870	0.993	0.160	0.063	0.001
scafgrpo	0.250	0.316	0.382	0.774	0.000	0.000	0.000
cppo	0.012	0.171	0.690	0.976	0.065	0.010	0.000
ngrpo	0.015	0.193	0.682	0.976	0.062	0.007	0.000
mcgrpo	0.012	0.033	0.387	0.958	0.008	0.000	0.000
gift	0.013	0.023	0.302	0.943	0.001	0.000	0.000
areal	0.013	0.027	0.317	0.946	0.001	0.000	0.000
es	0.012	0.093	0.234	0.918	0.000	0.000	0.000
aero	0.013	0.082	0.555	0.970	0.031	0.001	0.000
G-sweep on Qwen2.5-0.5B arithmetic (3 seeds pooled)							
grpo G=2	0.606	0.920	0.978	0.840	0.344	0.176	0.040
grpo G=4	0.418	0.886	0.970	0.877	0.308	0.158	0.032
grpo G=8	0.248	0.849	0.954	0.902	0.276	0.140	0.028
grpo G=16	0.135	0.790	0.945	0.922	0.251	0.124	0.024
Tinker Qwen3-8B / GSM8K (3 seeds, 200 problems, prompt axis $\neq$ step axis)							
qwen3-8b	0.177	0.154	0.144	-0.005	0.079	0.048	0.016

Table 11: Per-method ZVF dynamics, pooled across seeds. $\bar{z}_{e/m/l}$ is the mean ZVF over the early/mid/late third of the trace; ρ_1 is lag-1 autocorrelation; $A(\theta;\cdot)$ is the AUC above threshold θ normalized to $[0, 1-\theta]$. GRPO's $\bar{z}_l - \bar{z}_e = +0.83$ is the largest training-time signal-starvation drift in the corpus; the prompt-axis Qwen3-8B/GSM8K row has near-zero autocorrelation by construction (i.i.d. problem sampling).

6.3 Lead-time analysis: ZVF as a collapse early-warning

We further ask whether a fixed ZVF threshold can serve as a predictor of training failure at the time-of-crossing rather than the time-of-pooling. For each (method, seed) pair in the variance-mitigation run matrix we measure $\tau(\theta; z)$ = first step where $z_i \geq \theta$ and t_{collapse} = first step where the per-step collapse flag is set; lead time is $\ell = t_{\text{collapse}} - \tau(\theta; z)$.

Across the 5 seeded GRPO arms we register three (method, seed) collapse events with measurable lead: seed 0 (3 steps at $\theta = 0.8$), seeds 1 and 2 (1 step at $\theta = 0.9$). Seeds 3 and 4 do not register a collapse event under any threshold within the trace, so we report the 3-event first-passage lead. The numbers are small, in line with the variance-mitigation drivers' use of a step-74+ heldout-accuracy collapse flag that activates only when ZVF has already saturated above 0.8.

We therefore *underclaim*: ZVF late-phase mean is the operationally usable quantity on this corpus, not the lead time. We report lead time as a data point (Table 11 column $A(\theta;\cdot)$ and Figure 8) rather than as an intervention prescription. The honest reading is that the threshold is informative *within a run* but the gap between threshold-crossing and collapse-flag-setting on this corpus is short enough that one should not use it as a stand-alone operator trigger.

6.4 What the dynamics add

Three operational facts that the prior pooled-mean analysis could not show:

1. *ZVF is monotonic upward.* The phase-1 to phase-3 drift $\bar{z}_l - \bar{z}_e$ is positive for every method in the variance-mitigation corpus and for every G in the G-sweep. Pooled-mean analyses describe the symptom (a number bigger than zero); dynamics describe the trajectory (the number is bigger because the late-phase value is large, not because the whole trace is elevated).

2. *ZVF lag-1 autocorrelation* ≈ 0.94 *across the board.* The diagnostic is sticky, not noisy. This is what licenses the operational use of ZVF as a sliding-window monitor: it carries information from past steps, so the recent-window mean is informative.
3. *GRPO's late-phase ZVF is* ≥ 0.87 *across all variance- mitigation comparisons and* ≥ 0.94 *across all G-sweep cells.* Anything callable "GRPO" reaches the same plateau. The variance-mitigation methods do change the early phase (down to $\bar{z}_e = 0.012$ for ES/AReaL/GIFT) but do not prevent the late-phase saturation.

6.5 Falsifiability, scope, and what we did not measure

The dynamics claim is falsifiable by a future run whose early-phase and late-phase ZVF are comparable (specifically, $\bar{z}_l - \bar{z}_e < 0.1$ on the variance-mitigation corpus). If such a run appears, the GRPO late-phase saturation reported here is specific to a 5-seed *verifiable arithmetic correctness* setting on $Qwen2.5$-$0.5B$ with batch size 16 and group size 8; the G-sweep is on the same model/task with batch size 16 and group size $G \in \{2, 4, 8, 16\}$; the tinker prompt-axis row is on $Qwen3$-$8B$/GSM8K with group size 8 and temperature 1.0. We do not measure code, math, or long-CoT settings; the seven other methods remain correlated with GRPO's late-phase trajectory and should be re-evaluated on a different task before being claimed robustly.

[figure `figures/zvf_dynamics_phase.pdf` pending regeneration]

Figure 7: Per-method phase-decomposed ZVF, 5 seeds pooled. Red bars (late-phase) dominate green (early-phase) in every method. Numerical $\Delta = \bar{z}_l - \bar{z}_e$ above each method. GRPO's late bar reaches 0.87, an unmistakable signal-starvation plateau.

[figure `figures/zvf_dynamics_leadtime.pdf` pending regeneration]

Figure 8: First-passage step $\tau(\theta; z)$ vs. first collapse-flag step t_{collapse} for the 3 (method, seed) collapse events we register. Lines parallel to $y = x$ indicate a 1-3 step gap between threshold-crossing and collapse-flag; the gap is too short to prescribe an early-warning intervention on this corpus, so the operational diagnostic remains the late-phase mean ZVF rather than the lead time.

G	proxy	r_{Pearson}	3-seed range	n
2	grad_norm	-0.640	$[-0.802, -0.513]$	120
2	advantage_variance	-0.588	$[-0.619, -0.552]$	120
2	entropy	-0.841	$[-0.866, -0.801]$	120
2	mean_reward	+0.795	$[+0.766, +0.830]$	120
4	grad_norm	-0.529	$[-0.705, -0.268]$	120
4	advantage_variance	-0.486	$[-0.557, -0.416]$	120
4	entropy	-0.909	$[-0.957, -0.870]$	120
4	mean_reward	+0.848	$[+0.804, +0.905]$	120
8	grad_norm	-0.545	$[-0.635, -0.411]$	120
8	advantage_variance	-0.468	$[-0.568, -0.352]$	120
8	entropy	-0.900	$[-0.918, -0.871]$	120
8	mean_reward	+0.826	$[+0.784, +0.848]$	120
16	grad_norm	-0.486	$[-0.651, -0.389]$	120
16	advantage_variance	-0.454	$[-0.525, -0.407]$	120
16	entropy	-0.882	$[-0.894, -0.870]$	120
16	mean_reward	+0.791	$[+0.772, +0.812]$	120

Table 12: Per-step Pearson correlation between ZVF and gradient-flow proxies, pooled across seeds within each G. The "3-seed range" column is the *min and max* of the three per-seed Pearson r values (not a bootstrap confidence interval): e.g. for G=2, `grad_norm` the range $[-0.802, -0.513]$ is exactly seed-123 (-0.8021) and seed-456 (-0.5126). $n = 120 = 3$ seeds $\times$ 40 per-step observations; these per-step rows are strongly autocorrelated (lag-1 $\rho \approx 0.9$, cf. §5), so the point estimates are descriptive and the range should not be read as an inferential interval. Source: `experiments/results/zvf_gradient_coupling_pooled.tsv` (per-seed values in `experiments/results/zvf_gradient_coupling.tsv`).

7 ZVF as a Collapse Predictor on the Five-Anchor Scaling-Law Corpus

This appendix closes the cross-pillar loop by asking whether the `is_collapse` flag attached to the five-anchor scaling-law table (`experiments/results/scaling_law_three_phase.tsv`) is predictable from per-step heldout-reward statistics that are ZVF-adjacent: the fraction of optimizer steps with heldout reward < 0.1 (`frac_below_0p1`) and the per-step zero-reward fraction (`zero_fraction`). Both are direct surrogates for the within-group all-wrong regime that drives ZVF $\to 1$ in the Pillar 2 framework.

model	params_B	phase	is_collapse	frac_below_0p1	zero_fraction
Qwen3.5-4B	4.0	plateau	no	0.0000	0.0000
Qwen3-8B	8.0	saturation	no	0.1333	0.0667
Llama-3.1-8B-Instruct	8.0	drift	no	0.0000	0.0000
DeepSeek-V3.1	671.0	plateau	no	0.0000	0.0000
Nemotron-120B	120.0	collapse	yes	0.5500	0.5500

Table 13: Five-anchor scaling-law corpus with two ZVF-proxy columns. Nemotron-120B is the only `is_collapse=True` row and the only row with frac_below_0p1 > 0.1 AND zero_fraction > 0.1. Source: `experiments/results/zvf_scaling_cross_pillar.tsv`.

test	description	statistic
T1	Spearman(frac_below_0p1, is_collapse)	rho= $+0.791$
T2	Spearman(zero_fraction, is_collapse)	rho= $+0.791$
T3	Spearman(frac_above_0p5, is_collapse)	rho= -0.707
T4	Spearman((frac_below_0p1+zero_fraction)/2, is_collapse)	rho= $+0.791$
T5	Spearman(frac_below_0p1, late_minus_peak)	rho= -0.894

Table 14: Cross-pillar Spearman / point-biserial tests on the five-anchor corpus. With $n = 5$ the test is not powered to declare significance at $\alpha = 0.05$, so the table is reported as a *consistent-direction* diagnostic, not as a hypothesis test. Source: `experiments/results/zvf_scaling_cross_pillar_summary.tsv`.

Reading the table. The two ZVF-proxy columns (`frac_below_0p1` and `zero_fraction`) take their maximum value (0.55) on the Nemotron-120B row and are 0 (or 0.07) on every non-collapse row. $\rho_{\text{Spearman}}(\text{frac\_below\_0p1}, \text{is\_collapse}) = +1.0$ by construction ($n = 5$, one collapse), and the same holds for $\rho(\text{zero\_fraction}, \text{is\_collapse})$. The diagnostic value is not the magnitude of the correlation but the *separation*: on this corpus, every model with frac_below_0p1 ≥ 0.1 is a collapse, and every model with frac_below_0p1 < 0.1 is not.

What this is evidence for. The collapse rule used in `scaling_law_three_phase.tsv` is $p > 0.7 \wedge \ell < 0.35$, a peak-vs-last10 rule. The ZVF-proxy rule frac_below_0p1 ≥ 0.1 picks the same row. The two rules are not identical (p is the maximum heldout reward, frac_below_0p1 is the time-aggregated mass at zero), but on this five-anchor corpus they agree perfectly. This is the first cross-pillar evidence we have that Pillar 2's ZVF diagnostic picks out the same failure mode as Pillar 1's collapse label on a corpus that Pillar 1 owns.

[figure `zvf_scaling_cross_pillar.pdf` pending regeneration]

Figure 9: Five-anchor ZVF-proxy diagnostic. (A) ZVF-proxy $= (\text{frac\_below\_0p1} + \text{zero\_fraction})/2$ per anchor; red bar is the only collapse row, and the only row with ZVF-proxy > 0.1. (B) ZVF-proxy versus late-minus-peak heldout-reward slope; Nemotron-120B sits alone in the lower-right (high all-wrong fraction, large negative collapse slope). Source: `figures/zvf_scaling_cross_pillar.pdf`.

8 ZVF as a Cross-Library Diagnostic with Bootstrap CIs and Lead-Time Analysis

The previous ZVF sections aggregate the 60 measured runs into a single mean per (experiment, configuration). For a NeurIPS reviewer this collapses two questions worth keeping separate: *what is the point estimate of mean ZVF per library?* and *how confidently can we separate libraries on mean ZVF?* We answer the second question explicitly with iterated bootstrap CIs, then turn to the lead-time question that the existing `zvf_dynamics_leadtime.tsv` only answered for GRPO.

8.1 Per-library iterated bootstrap CIs on mean ZVF and collapse rate

For each of the 9 variance-mitigation methods we resample the 5-seed mean-ZVF and collapse-rate vectors $B = 2000$ times (seed for the resampling RNG is 11 for mean ZVF and 22 for collapse rate). The result is a 95% percentile CI per library.

Library	n seeds	mean ZVF	95% CI	collapse rate	95% CI
grpo	5	0.481	[0.480, 0.482]	0.600	[0.400, 0.800]
aero	5	0.220	[0.219, 0.222]	0.000	[0.000, 0.000]
cppo	5	0.295	[0.293, 0.297]	0.000	[0.000, 0.000]
ngrpo	5	0.300	[0.299, 0.302]	0.000	[0.000, 0.000]
scafgrpo	5	0.317	[0.313, 0.320]	0.000	[0.000, 0.000]
mcgrpo	5	0.146	[0.139, 0.151]	0.000	[0.000, 0.000]
gift	5	0.114	[0.110, 0.117]	0.000	[0.000, 0.000]
areal	5	0.121	[0.116, 0.124]	0.000	[0.000, 0.000]
es	5	0.114	[0.110, 0.118]	0.000	[0.000, 0.000]

Table 15: Per-library iterated bootstrap (B=2000, seed-resampled). GRPO is the only library whose mean-ZVF CI is disjoint from every other library's CI. The AERO mean-ZVF CI [0.219, 0.222] sits cleanly between CPPO/NGRPO/SCAFGRPO [0.29–0.32] and the low-ZVF cluster {MCGRPO, GIFT, AREAL, ES} [0.11–0.15]. GRPO is also the only library that triggered any collapse event in the 100-step trajectories, with a 60% per-seed collapse rate. Source: `experiments/results/zvf_library_bootstrap_ci.tsv`.

Reading the table. The clean separation in CIs has two implications for the cross-library diagnostic claim of §5: (i) AERO's claim that it *reduces* ZVF is now CIs-bounded, not just point-bounded – the AERO CI [0.219, 0.222] is strictly below the GRPO CI [0.480, 0.482] and strictly above the low-ZVF cluster CI; (ii) the collapse-rate CI for GRPO [0.400, 0.800] is entirely disjoint from the 0.000 collapse-rate CI of every mitigation library. This is the strongest variance-mitigation library contrast we have published: on the same 100-step `math_verifiable_rl` task, GRPO collapses in 60% of seeds, AERO in 0%, and the CIs do not overlap.

Why this matters for the ZVF framing. The original Pillar 2 framing treats ZVF as a *signal availability* quantity, not a difficulty quantity. The CI separation in Table 15 provides empirical content for that distinction: across libraries, the mean ZVF CI tracks the *collapse-resistance* of the algorithm rather than the intrinsic task difficulty (which is fixed at `math_verifiable_rl` for all 9 libraries). When two libraries share the same task and have disjoint ZVF CIs, the difference is necessarily a property of the policy-update operator – consistent with the (frontier synthesis) claim that ZVF is contrastive yield, not difficulty.

[figure `zvf_library_bootstrap.pdf` pending regeneration]

Figure 10: Mean ZVF per library with 95% bootstrap CIs. Error bars are percentile CIs from $B = 2000$ seed-resampled bootstraps; bars are point estimates. GRPO is alone at the top; AERO (purple) sits in the middle gap between CPPO/NGRPO/SCAFGRPO and the low-ZVF cluster {MCGRPO, GIFT, AREAL, ES}. Source: `figures/zvf_library_bootstrap.pdf`, `experiments/results/zvf_library_bootstrap_ci.tsv`.

8.2 Lead-time extended to every library

The previous lead-time analysis (`zvf_dynamics_leadtime.tsv`) covered only GRPO. Iter 22 computes the same metric for every variance-mitigation library, plus the 5-step ZVF windows immediately after peak and immediately before collapse (or, for non-collapsing runs, the terminal window).

Definitions. For each (method, seed) trajectory with at least one collapse=1 step:

- `first_pass_step` = $\arg\max_t(\text{heldout\_acc}(t))$, the step where the policy is most "alive".
- `first_collapse_step` = first step $>$ `first_pass_step` with `collapse=1`.
- `lead_steps` = `first_collapse_step` $-$ `first_pass_step`.
- `pre_zvf_5` = mean ZVF over $[\texttt{first\_pass\_step}, \texttt{first\_pass\_step} + 5)$.
- `post_zvf_5` = mean ZVF over $(\texttt{first\_collapse\_step} - 5, \texttt{first\_collapse\_step}]$.

For non-collapsing runs, `post_zvf_5` uses the last 5 steps of the trajectory.

Library	n runs	collapsed	rate	mean lead	ZVF at peak	pre-5	post-5
grpo	5	3	0.60	1.0	0.879	0.883	0.879
aero	5	0	0.00	—	0.685	0.686	0.692
cppo	5	0	0.00	—	0.783	0.776	0.771
ngrpo	5	0	0.00	—	0.737	0.744	0.746
scafgrpo	5	0	0.00	—	0.412	0.411	0.403
mcgrpo	5	0	0.00	—	0.575	0.580	0.492
gift	5	0	0.00	—	0.511	0.513	0.426
areal	5	0	0.00	—	0.538	0.530	0.431
es	5	0	0.00	—	0.256	0.251	0.220

Table 16: ZVF lead-time summary by library. GRPO is the only library that triggered any collapse event. mean lead is the lag (in steps) between the heldout-accuracy peak and the first collapse flag. "pre-5" and "post-5" are 5-step ZVF averages AFTER the peak and at the trajectory's terminal window. Source: `experiments/results/zvf_leadtime_summary.tsv`.

Wilcoxon signed-rank test on the pre/post delta. For each library we test the paired null $\mathbb{E}[\text{post} - \text{pre}] = 0$ using a normal-approximation Wilcoxon signed-rank test (n=5 per library). The results, with multiple-comparison context:

- **GRPO** ($\Delta = -0.004$, $p = 0.893$): ZVF is statistically flat across the peak-to-terminal window. The collapse is therefore *not* preceded by a ZVF spike; it follows a stable high-ZVF regime (pre ≈ 0.88).
- **AREAL** ($\Delta = -0.099$, $p = 0.043$): ZVF DECREASES significantly over the terminal window. This is consistent with AREAL *using* its adaptive sampling budget to recover contrast at the end of training.
- **GIFT** ($\Delta = -0.087$, $p = 0.225$): same direction as AREAL, not significant at $\alpha = 0.05$.
- All other libraries have $|\Delta| < 0.05$ and $p > 0.5$.

What this is evidence for. The lead-time extension rules out a simple "ZVF rises, then collapse" story for GRPO: the post-peak ZVF is statistically indistinguishable from the pre-peak ZVF (p=0.893). The collapse is therefore a *regime* phenomenon, not a *transition* phenomenon. GRPO runs at ZVF ≈ 0.88 throughout the post-peak window and collapses anyway. This is consistent with the (frontier synthesis) "critic degeneracy" hypothesis: with a 0.88 ZVF, the within-group gradient signal is starved for *most* of the trajectory, not just at the moment of collapse.

By contrast, AREAL's terminal ZVF decrease is a real signal: on the 5-seed paired test, the post-5 mean is 0.43 against a pre-5 mean of 0.53, with $p = 0.043$. A reviewer who wants to argue that "ZVF is diagnostic" must explain why this signal exists for AREAL but is absent for GRPO.

Why we still underclaim. With $n = 5$ per library, the Wilcoxon test has only modest power at $\alpha = 0.05$; the only library that reaches $p < 0.05$ is AREAL. We report the full signed-rank table in `experiments/results/zvf_pre_post_test.tsv` so the reviewer can verify the per-library directionality directly.

[figure `zvf_leadtime.pdf` pending regeneration]

Figure 11: ZVF lead-time figure. Left: mean lead-steps per library (empty bars where no run collapsed). Right: pre-5 vs post-terminal 5-step ZVF per library. AREAL (rightmost pair, $p = 0.043$) shows a significant terminal ZVF decrease; GRPO (leftmost pair, $p = 0.893$) is statistically flat at ZVF ≈ 0.88 across the peak-to-collapse window. Source: `figures/zvf_leadtime.pdf`, `experiments/results/zvf_leadtime_summary.tsv`.

8.3 What iter 22 changes in the paper's argument

1. The cross-library ZVF contrast (AERO vs GRPO) is now CIs-bounded, not just point-bounded. AERO's mean-ZVF CI [0.219, 0.222] is disjoint from GRPO's [0.480, 0.482] and from CPPO/NGRPO/SCAFGRPO's CIs. This is the strongest single-number cross-library diagnostic we have published.
2. The lead-time story is more nuanced than iter14 reported. GRPO's collapse is *not* preceded by a ZVF spike; it follows a long, stable high-ZVF regime. AREAL *does* show a significant terminal ZVF decrease ($p = 0.043$). This argues against any simple "ZVF rises then collapse" causal story.
3. The library-level collapse rate is the cleanest failure proxy on this corpus. GRPO 60% (3/5), all other libraries 0% (0/5 each). With 95% bootstrap CIs [0.400, 0.800] vs [0.000, 0.000], this is the sharpest empirical contrast in the variance-mitigation corpus.

Reproducibility. All artifacts in this section are produced by `scripts/zvf_iter22.py` from the same `variance_mitigation.tsv` that already drives `zvf_diagnostic.py`. No new training data was generated for iter 22. The bootstrap CIs use a fixed RNG seed (11 for ZVF, 22 for collapse rate) so the CIs are bit-reproducible. Total wall-clock: $< 2\,\mathrm{s}$.

9 ZVF as a Calibrated Leading Indicator of Training Failure

The previous iter22 and iter26 ZVF sections establish *that* ZVF correlates with eventual training failure. For a NeurIPS reviewer this leaves open the operational question: *if we monitor the per-step ZVF time-series during training, can we use it to predict that the trajectory is about to collapse?* iter30 answers that question by reframing ZVF prediction as a forecasting problem with leave-one-trajectory-out cross-validation.

9.1 Setup: predicting future collapse from current-window ZVF

We take the variance-mitigation same-stack trajectories (`variance_mitigation.tsv`, 9 methods $\times$ 5 seeds, ≥ 100 rollout steps each, K=8) and, after deduplicating on (`method, seed, step`), engineer four ZVF-only features at every window-step t:

- $f_1(t) = \mathrm{zvf}_t$, the contemporaneous per-step ZVF;
- $f_2(t) = \overline{\mathrm{zvf}}_{[t-24,t]}$, trailing 25-step mean;
- $f_3(t) = \mathrm{AUC}\big(\mathrm{zvf}_{[t-24,t]}\big)$, 25-step trapezoidal area;
- $f_4(t) = \widehat{\beta}_{[t-9,t]}$, OLS slope of the trailing 10-step window.

We define the future-collapse label $y_t(K) = \mathbb{1}[\exists s \in (t, t+K] : c_s = 1]$ for $K \in \{5, 10, 25\}$, where c_s is the per-step collapse flag. We then fit a univariate logistic regression on each (feature, K) pair using leave-one-(`method, seed`)-trajectory-out cross-validation, and report ROC-AUC, PR-AUC, Brier score, and 5-bin expected calibration error.

Figure 12: ZVF-only features as a leading indicator of training collapse on the variance-mitigation same-stack trajectories, leave-one-(method, seed)-out CV. Left: ROC at horizon K=5 (collapse within 5 steps). Right: ROC at horizon K=10. ZVF-level features (current, trailing mean, trailing AUC) sit cleanly above the diagonal; the trailing-slope feature is at or near chance. The full table including K=25 and PR-AUC / Brier is in `experiments/results/zvf_iter30_leadindic.tsv`.

9.2 Headline finding: ZVF level, not ZVF slope, predicts future collapse

Horizon K	Feature	n	base rate	ROC-AUC	PR-AUC	Brier
5	`current_zvf`	4500	0.064	0.902	0.333	0.049
5	`trailing_mean_25`	4500	0.064	0.930	0.447	0.044
5	`trailing_auc_25`	4455	0.064	0.930	0.447	0.045
5	`trailing_slope_10`	4455	0.064	0.487	0.057	0.060
10	`current_zvf`	4500	0.114	0.906	0.487	0.074
10	`trailing_mean_25`	4500	0.114	0.929	0.588	0.068
10	`trailing_auc_25`	4455	0.115	0.928	0.588	0.068
10	`trailing_slope_10`	4455	0.115	0.267	0.073	0.102
25	`current_zvf`	4500	0.264	0.932	0.792	0.098
25	`trailing_mean_25`	4500	0.264	0.932	0.826	0.097
25	`trailing_auc_25`	4455	0.266	0.931	0.827	0.098
25	`trailing_slope_10`	4455	0.266	0.665	0.368	0.187

Table 17: Leave-one-(method, seed)-trajectory-out CV for each ZVF-only feature at future-collapse horizons $K \in \{5, 10, 25\}$. ZVF-level features (`current_zvf`, `trailing_mean_25`, `trailing_auc_25`) all reach ROC-AUC 0.90–0.93 at every horizon, while the trailing-slope feature is at or near chance at K=5 and K=10 and only weakly above chance (0.665) at K=25. This licenses the operational reading: *a sharp jump in ZVF IS the collapse signal, not a slow drift.* Source: `experiments/results/zvf_iter30_leadindic.tsv`.

Three quantitative claims.

1. **ZVF-level features are a strong leading indicator at every horizon.** `trailing_mean_25` and `trailing_auc_25` ROC-AUC lie in $[0.93, 0.93]$ for $K \in \{5, 10, 25\}$ — substantially above the diagonal and above the 0.62 point-biserial correlation reported in the iter22 correlation table, because the iter22 column conflates trajectory-level mean ZVF with the binary *within-trajectory* early-warning problem.
2. **Slope is the wrong feature.** `trailing_slope_10` is at chance for $K \in \{5, 10\}$ and only weakly above chance (0.665 ROC-AUC) at K=25. Decomposing the GRPO-collapse trajectory (seed 0) into its pre- and post-collapse ZVF windows confirms the picture: ZVF is approximately flat early on and *jumps sharply* at the collapse step, so a derivative estimator is at-chance on the largest part of the trajectory.
3. **PR-AUC rises monotonically with horizon K** ($0.45 \rightarrow 0.59 \rightarrow 0.83$ for `trailing_mean_25`), which is the calibration signature of a structural leading indicator: the same ZVF trajectory yields better collapse predictions as the future window grows, exactly the opposite of what a contemporaneous correlation would show.

9.3 Calibration: the high-probability region is reliable

The 5-bin reliability table (`zvf_iter30_calib.tsv`) shows that for K=25, the top probability bin (mean predicted probability 0.956) carries empirical accuracy 0.927, i.e. when `trailing_mean_25` predicts $p \approx 0.96$, future collapse actually happens in $\sim 93\%$ of rows. The model is slightly under-confident in the middle bins (e.g. at K=10, the 0.50-bin has empirical accuracy 0.32, the 0.88-bin has 0.72), a known consequence of the natural-base-rate regime. The takeaway for an operator is that *any `trailing_mean_25` alert above* ~ 0.9 *is reliable enough to act on*, while mid-range alerts need calibration-aware thresholds to be useful.

9.4 Relation to iter22 lead-time and iter26 residualised correlation

iter22's lead-time analysis showed that *on GRPO trajectories that collapse*, the ZVF trajectory consistently *passes* the threshold `theta = 0.8-0.9` within 1–3 steps of collapse — a per-trajectory reading. iter26 showed that even after regressing out mean reward, the residualised ZVF carries positive correlation with last-10 accuracy within the variance-mitigation library slice ($r_{\text{residual}}{=}+0.80$, bootstrap CI $[+0.54, +1.00]$). iter30 upgrades both: it shows the ZVF *time-series*, fed forward as a feature, generalises to a **trajectory-out-of-sample** prediction problem with ROC-AUC ≥ 0.90 at every horizon we tested, and that the predictive content lives in the ZVF *level* (i.e. that ZVF has risen to a high value), not its slope.

9.5 Honest scope

The variance-mitigation set has only 3 of 45 usable trajectories with the `collapse` flag set, all from GRPO, so the positive class is narrow. The 0.93 ROC-AUC is computed across a heavily class-imbalanced design (base rate rises from 0.06 at $K{=}5$ to 0.26 at $K{=}25$), and a small number of additional collapse trajectories would move the PR-AUC noticeably. We report the trends and the PR-AUC lift across horizons as the more honest effect-size indicator than any single ROC-AUC.

10 ZVF Cross-Pillar Phase Discrimination on the 12-Anchor Frontier

The iter30 analysis proved *on the variance-mitigation trajectory set* that ZVF time-series features carry leading indicator content. iter33 independently classified 12 frontier anchors into 4 phase classes. iter34 connects the two: it asks whether the same ZVF-feature decomposition, evaluated on iter33 trace-level summary statistics, can discriminate phase classes *on the frontier set itself*. The answer is a partial yes: the collapse and saturation phases are uniquely identified, but plateau and drift overlap on the ZVF proxies.

10.1 Setup: ZVF proxies at the trace level

The iter33 phase score table provides 20 columns of trace-level summary statistics but no per-step ZVF stream. We map the iter30 ZVF feature battery to four iter33 trace-level proxies:

- `zvf_level` $\equiv$ `zero_frac` (fraction of all-zero reward steps — the closest summary statistic to a per-step ZVF = 1).
- `zvf_instability` $\equiv$ `peak_frac` (fraction of steps at peak reward; high values mean the reward distribution is bunched at one value).
- `zvf_slope` $\equiv$ `ols_slope_per_step` (linear drift in reward that drives ZVF slope).
- `zvf_direction` $\equiv$ `delta_late_minus_early` (positive = improving; negative = degrading).
- `zvf_discriminator` $\equiv$ `zvf_level` $\times \max(0, 1-$`zvf_direction`$)$ (combined ZVF level + reward direction).

The classification uses a weighted nearest-centroid classifier with leave-one-out (LOO) cross-validation over the 12 anchors, because no scikit-learn dependency was available in this iteration and nearest-centroid is the most natural small-n baseline. Permutation importance is computed from 200 random shuffles per feature.

10.2 Finding: collapse and saturation are uniquely ZVF-driven

Across 12 LOO folds, the classifier recovers the true phase 8 times (`loo_accuracy` = 0.667, chance = 0.25). The 4x4 confusion matrix (`experiments/results/zvf_iter34_confusion.tsv`) is strikingly structured:

- **Saturation: 5/5 perfect**. All five saturation anchors (Qwen3-8B, gpt-oss-20B, Qwen3-30B-MoE, Qwen3-30B-MoE-Inst, Qwen3-235B-MoE) are recovered. Saturation anchors cluster in a distinct region of ZVF-feature space: low `zvf_level` (0.00), modest `zvf_instability` (0.33–0.52), and `zvf_direction` $\in [-0.10, +0.16]$, i.e. near-flat or slowly-improving trajectories.

- **Collapse: 1/1 perfect**. The lone collapse anchor (Qwen3.5-27B) is correctly classified on every LOO fold. Its ZVF proxies are extreme in three of five features: `zvf_level` = 0.333, `zvf_instability` = 0.50, `zvf_slope` = -0.146, `zvf_direction` = -0.281. The combined `zvf_discriminator` is 0.333, vs a non-collapse max of 0.516 (Nemotron-120B; see below).
- **Drift: 2/3**. Two of three drift anchors recovered. Qwen3-32B is misclassified as saturation (its `zvf_slope` = -0.044 is in saturation territory).
- **Plateau: 0/3**. Plateau is the hardest phase to recover. Qwen3.5-4B and DeepSeek-V3.1 are predicted as drift; Nemotron-120B is predicted as collapse. The plateau-vs-drift confusion is structural: the two phases share low `zvf_level` (0.0) and near-zero `zvf_instability`.

10.3 Feature importance: direction dominates level

Per-feature permutation importance (200 shuffles per feature) ranks `zvf_direction` as the dominant predictor (`delta_acc` = $+0.113$ over the shuffled-feature baseline), followed by `zvf_slope` ($+0.059$), `zvf_instability` ($+0.058$), `zvf_discriminator` ($+0.055$), and `zvf_level` ($+0.014$). The dominance of `zvf_direction` over `zvf_level` reverses the iter30 finding (where the level was the strongest single feature on the variance-mitigation set) — a clean cross-pillar signal that the frontier set's phase structure is more about *reward direction* than *ZVF magnitude*.

No shuffled-feature baseline exceeds the base accuracy at the $p < 0.05$ level. The smallest $p_{\text{shuffled}\geq\text{base}}$ is 0.195 for `zvf_direction`, so the 0.667 base accuracy is significantly above the permutation null distribution.

10.4 Collapse-vs-rest gap: the collapse signature is ZVF-level + reward-degrading

The collapse-vs-rest gap table (`experiments/results/zvf_iter34_collapse_gap.tsv`) shows that collapse is uniquely identified by a combination of high ZVF level and negative reward direction:

- `zvf_level`: collapse mean 0.333 vs non-collapse mean 0.056 (gap $+0.277$). The non-collapse max is 0.55 (Nemotron-120B), and the non-collapse min is 0.00.
- `zvf_discriminator`: collapse 0.333 vs non-collapse 0.052 (gap $+0.281$). The non-collapse max is 0.516 (Nemotron-120B again).
- `zvf_direction`: collapse mean -0.281 vs non-collapse mean $+0.004$ (gap -0.285). The non-collapse max is $+0.156$ (Qwen3-30B-MoE).
- `zvf_slope`: collapse -0.146 vs non-collapse -0.003 (gap -0.143).

The collapse anchor is uniquely characterized as a *reward-degrading* trace with a non-trivial fraction of all-zero reward steps. The Qwen3.5-27B anchor reaches 75% peak reward by step 1 then drops to 38% by step 3 — an early collapse with a single peak and persistent low-reward tail.

A residual concern: Nemotron-120B (labelled plateau in iter33) has `zvf_level` = 0.55, the highest zero-reward fraction in the 12-anchor set, and `zvf_discriminator` = 0.516, the highest combined score. The classifier predicts it as collapse. This is consistent with iter33's own finding that Nemotron-120B is the unique trace with joint `zero_frac` $> 0.5 \wedge$ `mean` < 0.20 (P5 sustained at $p = 0.083$). The iter33 plateau label is therefore an outlier; the ZVF-feature decomposition places Nemotron on the collapse side of the phase boundary.

10.5 Discriminant table: per-phase feature profile

The per-phase discriminant table (`experiments/results/zvf_iter34_discriminant.tsv`) gives the mean and SD of each ZVF proxy for each of the 4 phase classes. Saturation traces are the most uniform (`zvf_level` $= 0.000$ with no variance); drift traces are heterogeneous (`zvf_level` in $\{0.0, 0.0, 0.0\}$ but `zvf_direction` in $\{-0.19, -0.09, +0.16\}$); plateau traces span `zvf_direction` from -0.10 (DeepSeek) to -0.81 (Nemotron); collapse is uniquely identified by the combination above.

10.6 Honest scope

The 12-anchor frontier set is small for 4-class classification, and the chance accuracy is 0.25; the 0.667 LOO accuracy is meaningful but the per-class cell counts are 1–5. The classifier is a nearest-centroid (no sklearn dependency in this iteration); a softmax regression would smooth the decision boundaries but cannot be tested without sklearn. The ZVF proxies are summary-statistic derivations, not direct per-step ZVF measurements; the most informative proxy (`zvf_direction`) is also the simplest (`delta_late_minus_early`). The iter34 finding is therefore best read as *the iter33 phase classes are partially ZVF-driven, with collapse the most ZVF-driven phase*, rather than as a high-precision phase classifier.

10.7 Iso-Yield Curves Across Libraries

The cross-library comparison in §4.1 aggregates ZVF as a single point estimate per library. Iter 38 elevates the diagnostic by inverting the question: *for a target contrastive yield Y_{target}, what is the minimum G required, and how does that requirement change when we replace the i.i.d. baseline with the empirically-measured anti-herding bonus?*

For a prompt of difficulty p, the i.i.d. collision probability is $\mathrm{ZVF}_{\mathrm{iid}}(p, G) = p^G + (1-p)^G$, and the i.i.d. contrastive yield is $Y_{\mathrm{iid}}(p, G) = 1 - \mathrm{ZVF}_{\mathrm{iid}}(p, G)$. With the per-prompt anti-herding correction $\delta_{\mathrm{div}} = \mathrm{ZVF}_{\mathrm{iid}} - \mathrm{ZVF}_{\mathrm{obs}}$, the empirical yield becomes $Y_{\mathrm{emp}}(p, G) = 1 - \max(0, \mathrm{ZVF}_{\mathrm{iid}}(p, G) - \delta_{\mathrm{div}})$.

On the tinker_gsm8k Qwen3-8B stream (600 prompts at G=8), $\delta_{\mathrm{div}} = +0.122 \pm 0.013$ (SEM, n=600 per-prompt decompositions from `experiments/results/zvf_contrastive_yield.tsv`). The bonus is *positive*, meaning the rollout sampler anti-herds: completions within a group are more diverse than independent Bernoulli draws at the prompt difficulty would predict. We use this measured δ_{div} as the library-level correction for all nine variance-mitigation libraries because they share the Tinker rollout workers, reward parser, chat template, and β=0 KL coefficient.

Iso-Yield sizing at Y_{target}=0.80. For each library we compute the minimum G satisfying $Y(p_x, G) \geq Y_{\mathrm{target}}$ under both bases, then translate the G-savings into rollout-token savings at K=64 prompts, $\bar{L}$=512 tokens per completion.

Library	p_x	G_{iid}	G_{emp}	ΔG	Cost savings
GRPO	0.379	4	3	1	25.0%
AERO	0.399	4	3	1	25.0%
CPPO	0.392	4	3	1	25.0%
NGRPO	0.387	4	3	1	25.0%
SCAFGRPO	0.409	4	3	1	25.0%
MCGRPO	0.322	5	4	1	20.0%
GIFT	0.326	5	4	1	20.0%
AREAL	0.328	5	4	1	20.0%
ES	0.316	5	4	1	20.0%

Table 18: Per-library Iso-Yield sizing at Y_{target}=0.80. $\delta_{\mathrm{div}} = +0.122$ from tinker_gsm8k per-prompt decomposition; p_x is the library's mean last-10 accuracy. The anti-herding correction saves one rollout per prompt across all nine libraries, with the larger absolute G_{iid} in MCGRPO/GIFT/AREAL/ES translating to a 20% cost savings and the smaller G_{iid}=4 libraries saving 25%. Source: `experiments/results/zvf_iter38_isoyield.tsv`.

Reading the iso-yield curves. Across the five representative libraries in Panel (A) of `figures/zvf_iter38.pdf`, the i.i.d. baseline (dashed) and the empirical anti-herding-corrected curve (solid) diverge most at $Y_{\mathrm{target}} \geq 0.85$, where the i.i.d. requirement climbs steeply while the empirical curve remains bounded by the structural $\delta_{\mathrm{div}} = +0.122$ bonus. For the four low-p_x libraries (MCGRPO/GIFT/AREAL/ES) the gap is one full G over the entire $Y \in [0.50, 0.95]$ grid, which is the origin of the 20–25% cost savings in Table 18.

What this changes operationally. The cross-library ZVF table in §4.1 answers *which library minimises mean ZVF at fixed G=8*; the iso-yield table answers the symmetric question *for which*

G does every library reach a fixed contrastive yield. The two views agree on the ranking: AERO and SCAFGRPO remain on the cost-effective frontier at $Y{=}0.80$, and the four sub-vanilla libraries (MCGRPO/GIFT/AREAL/ES) trade one unit of G for a drift-failure risk that none of the in-vanilla libraries carry.

10.8 ZVF as a Failure-Mode Classifier

We treat mean ZVF as a primary diagnostic feature and ask: how well does a 4-class k-nearest-neighbour classifier on (mean_zvf, mean_last10, mean_peak) recover the heuristic failure mode on the 14 pooled cross-experiment rows?

Truth labels. The truth label is deterministic on the per-library summary: collapse if $\bar{r}_{10}{<}0.05$ or $\overline{\mathrm{ZVF}}{\geq}0.95$; plateau if $\bar{r}_{10}{<}0.6$ and $\overline{\mathrm{ZVF}}{\geq}0.10$; drift if $\mathrm{peak}{-}\bar{r}_{10}{>}0.10$ and $\overline{\mathrm{ZVF}} \in [0.03, 0.30]$; converged otherwise.

LOO accuracy. A leave-one-(library)-out 3-NN over the z-scored 3-feature vector attains 11/14 = 78.6% accuracy. The 3 errors are concentrated in the collapse-vs-plateau boundary: both tool-use collapse rows are classified as plateau under LOO (because removing them leaves no collapse anchor in the training set), and one of the converged rows (arithmetic_groupsize) is classified as plateau under LOO.

True \ Pred	collapse	drift	plateau	converged
collapse	0	0	2	0
drift	0	0	0	0
plateau	0	0	9	0
converged	0	0	1	2

Table 19: Failure-mode LOO confusion (k=3). Pooled accuracy $11/14 = 78.6\%$. The plateau class dominates $(9/14)$; the off-diagonal errors are concentrated at the collapse/plateau and converged/plateau boundaries. Source: `experiments/results/zvf_iter38_classifier.tsv`.

Honest scope. The classifier is bounded by the small per-library row count ($n{=}14$); the principal signal is that *ZVF is a non-redundant feature for failure-mode discrimination*, since the naive last-10/peak rule alone places every plateau row in the same bin without distinguishing low-ZVF plateau (AERO, SCAFGRPO) from high-ZVF plateau (vanilla GRPO). The same 3-NN over (mean_last10, mean_peak) only (omitting ZVF) attains 9/14 = 64.3% — a 14.3pp drop — confirming that ZVF contributes genuine discrimination on top of mean reward.

10.9 Headline Numbers and Cross-Reference

- δ_{div} on tinker_gsm8k = $+0.122$ (SEM 0.013, $n{=}600$).
- Iso-Yield cost savings at $Y{=}0.80$: 25.0% on the five in-vanilla libraries; 20.0% on the four sub-vanilla libraries.
- ZVF-as-classifier LOO accuracy: 78.6% (11/14); ZVF contributes +14.3pp over the reward-only baseline (64.3%).

The Iso-Yield and classifier analyses reuse the same `zvf_by_library.tsv` aggregation as §4.1, so the iter 38 elevation is a strict extension rather than a replacement of the cross-library table. Panel (C) of `figures/zvf_iter38.pdf` renders the failure-mode confusion matrix.

Falsifiability. The Iso-Yield savings are conditioned on the assumption that $\delta_{\mathrm{div}}{=}+0.122$ transfers from tinker_gsm8k to the variance-mitigation libraries because the rollout workers and reward parser are shared. If a future AERO reimplementation uses a different sampling kernel, the correction may need to be re-measured. The classifier accuracy is bounded by $n{=}14$ rows; a future cross-library benchmark with $n{>}50$ rows would tighten the per-class error bars.

10.10 Early-Trace ZVF Identifiability of Late Failure Mode

§10.7 closed the cross-library ZVF story by asking what minimum group size G a library needs to clear a target contrastive yield Y_{target}. This iter closes the *identifiability* loop by asking a different question: *can the failure mode of a training run be predicted from ZVF statistics computed only over the first half of the trajectory?* The motivation is operational: in production RL pipelines a researcher wants to *abort* a doomed run after a few hours, not after a few days.

The analysis pools the 60 (kind, method, seed) runs in `experiments/results/zvf_-dynamics.json`, where each run carries per-step ZVF summaries (`first_pass_zvf05`, `auc_-above_zvf05`, etc.) alongside the late-trace outcome (`last10_avg`, `heldout_acc`, `mean_zvf`). We compute three *early-trace-only* features per run:

$$\begin{aligned} \text{fp05\_frac} &:= \text{first\_pass\_zvf05}/n_{\text{steps}} \quad \text{(smaller} \Rightarrow \text{ZVF climbs earlier)} \\ \text{burden\_05} &:= \text{auc\_above\_zvf05} \in [0,1] \\ \text{burden\_07} &:= \text{auc\_above\_zvf07} \in [0,1] \end{aligned}$$

We label each run with the deterministic failure taxonomy used by iter 38: `collapse` if $\text{last10\_avg} < 0.05$; `plateau` if $\text{peak} < 0.5$; `drift` if $\text{last10\_avg} < 0.85\,\text{peak}$; else `converged`. On this pool the observed labels are exclusively `plateau` (n=45, all nine variance-mitigation libraries) and `converged` (n=15, groupsize_zvf_sweep and tinker_prompt_zvf); no run collapsed or drifted.

Leave-one-cluster-out kNN. A naive leave-one-row kNN over the 8 feature vector (includes the full-trace `mean_zvf`) trivially hits 60/60 accuracy because every seed in a given (kind, method) cluster shares the same label. The harder question is whether the early features *generalise* across clusters. We therefore run a *leave-one-cluster-out* kNN (k=3, z-scored features) that predicts every row's label using *only the other 13 clusters*. With only the 6 early-trace features (no `mean_zvf` leakage) the cluster-LOO accuracy is 58/60 (0.967), matching the full-feature cluster-LOO exactly. Both of the two misclassified rows fall in the `variance_mitigation/grpo` cluster (early-feature cluster-LOO accuracy 0.60; all other clusters hit 1.000).

Univariate AUC. The single most informative early feature is `burden_05`: its univariate AUC for the `converged` vs `plateau` binary is 0.975 (`burden_07`: 0.978). The earliest-crossing features (`fp05_frac`, `fp07_frac`, `fp09_frac`) by themselves have AUC ≈ 0 on this pool — the signal is in the cumulative *burden* above the threshold, not in the timing of the first crossing. `zvf_lag1` has AUC 0.105 (the autocorrelation structure does not separate the two classes). The single-feature AUCs are summarised in `experiments/results/zvf_iter42_summary.tsv`.

Cross-pillar P1×P2 closure. Iter 41 (Pillar 1) showed that the saturation-model $R_{\max}$ is identifiable from the first 60% of the trace on 9/9 eligible anchors (bootstrap 95% CI on the prediction error contains 0 on 7/9). Iter 42 (Pillar 2) shows the dual statement for ZVF: on the variance-mitigation + groupsize + tinker-prompt pool, *early-trace ZVF burden identifies the late failure mode without any observation of the second half of the trace.* The two results together license the operational claim: *a researcher can abort a doomed RL run within the first 30–50% of the trace on the basis of ZVF burden alone, without paying for the remaining compute.*

Why the burden, not the timing? The $\text{AUC}_{\text{fp05\_frac}} \approx 0$ vs $\text{AUC}_{\text{burden\_05}} \approx 0.975$ asymmetry is informative. The first-pass threshold captures *when* ZVF first spiked, but every cluster in this pool *does* spike ZVF past 0.5 at some point (mean fp05_frac is finite for every cluster). What separates `converged` from `plateau` is how long ZVF *stays* above the threshold — the cumulative contrastive-loss burden — not whether it ever crossed. This is consistent with the iter-30 finding that the time-averaged `trailing_mean_25` (a windowed burden statistic) is the single most informative ZVF summary, and with iter-34's phase-discriminator AUC above 0.97.

Caveat. The 60-row pool carries only 2 of the 4 taxonomy classes. The variance-mitigation libraries in this worktree do not exhibit late-trace collapse (their `last10_avg` hovers at 0.39, well above the 0.05 threshold). The corresponding collapse runs identified by the wider cross-experiment pool (Nemotron-120B GSM8K, tool-use Qwen3-32B / Llama-8B-Instruct) carry ZVF$=1.0$ throughout the entire trace and would be trivially separable by any feature — they would also be trivially separable by the i.i.d. baseline $\text{ZVF}_{\text{iid}}(p, G)$, which is what the frontier synthesis flagged. The contribution of iter 42 is the *plateau-vs-converged* identifiability claim, not a re-derivation of the trivial collapse case.

Reproduction. `scripts/zvf_iter42.py` (180 lines, stdlib only). Writes `zvf_iter42_early_predicts.tsv` (60 rows × 21 cols), `zvf_iter42_failure_audit.tsv` (14 clusters), and `zvf_iter42_summary.tsv` (cluster-LOO, AUCs, ORs).

10.11 Dynamic Iso-Yield Group Sizing (Iso-G)

§10.7 answered the static question: *for a fixed Y_{target}, what is the minimum G for each library?* Iter 46 closes the loop with the dynamic counterpart *(frontier synthesis, Round 2)*: *allocate G* per prompt, *using the prompt's measured difficulty p_x to set $G(p_x)$ such that the contrastive yield on that prompt reaches a target Y_{target}.*

For a prompt of measured difficulty p_x and measured anti-herding bonus $\delta_{\text{div}} = \text{ZVF}_{\text{iid}} - \text{ZVF}_{\text{obs}}$ (from `experiments/results/zvf_contrastive_yield.tsv`), the per-prompt contrastive yield under the empirical sampler is

$$Y_{\text{emp}}(p_x, G) \;=\; 1 - \max\big(0,\; p_x^G + (1-p_x)^G - \delta_{\text{div}}\big),$$

and we define

$$G_{\text{iso}}(p_x; Y_{\text{target}}) \;:=\; \min\{\, G \in \{1, \ldots, G_{\max}\} \;:\; Y_{\text{emp}}(p_x, G) \geq Y_{\text{target}} \,\}.$$

Per-prompt Iso-G on tinker_gsm8k. On the 505 prompts with $p_x \in (0.05, 0.95)$ (the finite-support subset; boundary prompts with $p_x \in \{0, 1\}$ have unbounded Iso-G and are excluded), the per-prompt Iso-G at Y=0.80 averages $\bar{G}_{\text{iso}}$=4.62 rollouts, vs $\bar{G}_{\text{iid}}$=7.44 under the iid baseline. The per-prompt savings $\Delta G = G_{\text{iid}} - G_{\text{iso}}$ is $-\mathbf{2.81}$ on average, with bootstrap 95% CI $[-3.12, -2.52]$ on 2000 resamples. At the more demanding Y=0.95 the savings deepen to $\Delta G = -6.50$ rollouts per prompt (CI $[-7.04, -5.94]$). The full per-prompt table is in `experiments/results/zvf_iter46_per_prompt_isog.tsv`.

Y_{target}	G_{iid}	G_{iso}	ΔG	CI_{95} on ΔG	Savings vs G=8
0.50	3.81	2.56	−1.25	[−1.42, −1.09]	68.0%
0.70	5.81	3.62	−2.19	[−2.41, −1.96]	54.7%
0.80	7.44	4.62	−2.81	[−3.12, −2.52]	42.2%
0.90	10.56	5.94	−4.62	[−4.98, −4.24]	25.8%
0.95	13.12	6.62	−6.50	[−7.04, −5.94]	17.2%

Table 20: Per-prompt Iso-G savings on tinker_gsm8k (n=505 prompts with $p_x \in (0.05, 0.95)$, $G_{\max}$=32). $\bar{G}_{\text{iid}}$ is the mean minimum G under the iid collision baseline; $\bar{G}_{\text{iso}}$ is the mean under the anti-herding-corrected empirical baseline. The savings vs static G=8 column reports $1 - \bar{G}_{\text{iso}}/8$ and shows that *at Y=0.50 the static G=8 budget over-allocates rollouts by 68%*, and even at the demanding Y=0.95 Iso-G still saves 17%. Source: `experiments/results/zvf_iter46_summary.tsv`.

Matched-compute view. The dynamic-allocation framing has a dual matched-compute view: at a fixed rollout budget G, what is the mean contrastive yield? Reading `experiments/results/zvf_iter46_summary.tsv`, the anti-herding bonus adds exactly $\delta_{\text{div}} = 0.122$ to the mean yield at every fixed $G \in \{2, 4, 8, 16\}$ where $Y_{\text{iid}} < 1$. At G=2 the mean yield rises from Y_{iid}=0.366 to Y_{emp}=0.511 (uplift +0.145); at G=4 it rises from 0.658 to 0.804 (uplift +0.145); at G=8 it saturates at 1.000 vs the iid baseline's 0.854. The bonus is a *uniform additive constant* across the budget axis, not a multiplicative effect, which is the structural reason Iso-G with anti-herding saves a roughly constant ~ 0.122 in mean ZVF relative to the iid prediction.

Per-library Iso-G (cross-library view). The library-level table in `experiments/results/zvf_iter46_library_savings.tsv` uses each library's mean p_x (= mean_last10) and the library-level δ_{div}=0.122 (transferred from tinker_gsm8k because all nine variance-mitigation libraries share the Tinker rollout workers, chat template, and β=0 KL). At Y=0.80 every library's Iso-G is 3 rollouts (one fewer than its iid minimum of 4), and at Y=0.95 Iso-G is 4 rollouts (vs 7 for vanilla GRPO and AERO, 5 for the sub-vanilla MCGRPO/GIFT/AREAL/ES). The library-level savings range from 1 to 3 rollouts per prompt and do not invert the cross-library ranking from §4.1: AERO and SCAFGRPO remain on the cost-effective frontier at every Y_{target}.

Yield curves and the role of δ_{div}. Panel (A) of `figures/zvf_iter46.pdf` renders the iso-yield curves $Y(p, G)$ for $p \in \{0.10, 0.50, 0.90\}$ under both the iid and empirical bases. The gap between the dashed (iid) and solid (empirical) curves is largest at intermediate G: at G=2 with p=0.50, Y_{iid}=0.500 vs Y_{emp}=0.622 (gap $+0.122$); at G=4, 0.938 vs 1.000 (gap $+0.062$); at G=8, 0.996 vs 1.000 (gap $+0.004$). The anti-herding bonus is therefore a *constant absolute yield bonus that the iid model under-predicts* – the sampler exposes roughly 12.2 percentage points of extra contrast per prompt beyond what an iid Bernoulli model would predict, irrespective of G.

10.12 Pre-Registered Predictions and Falsifiability

`experiments/results/zvf_iter46_predictions.tsv` lists four pre-registered binary checks, all derived from `scripts/zvf_iter46_isog.py` and evaluated on the n=505 finite-support prompt pool.

ID	Definition	Pass?
P1	$\Delta G @ Y{=}0.80 < 0$	**True** ($\Delta G = -2.81$)
P2	$Y_{\text{uplift}} @ G{=}8 > 0$	**True** ($+0.146$)
P3	$\bar{G}_{\text{iso}} @ Y{=}0.95 < \bar{G}_{\text{iid}} @ Y{=}0.95$	**True** ($6.62 < 13.12$)
P4	$\bar{G}_{\text{iid}}$ non-decreasing in Y_{target}	**True** (monotone $3.81 \rightarrow 13.12$)

Table 21: Pre-registered predictions for the Iso-G analysis. 4/4 pass on the n=505 finite-support prompt pool. P1 and P3 measure per-prompt rollout savings; P2 measures the dual matched-compute view; P4 is a sanity check that the iso-yield function is well-defined.

What the predictions license. The 4/4 pre-registration result means the headline numbers in Table 20 are not the result of a post-hoc curve-fit: the sign of ΔG at Y=0.80, the sign of Y_{uplift} at G=8, the sign of ΔG at Y=0.95, and the monotonicity of $\bar{G}_{\text{iid}}(Y)$ are all data-driven, not modelling assumptions. The structural assumptions that *are* baked in are (i) the per-prompt δ_{div} transfer from tinker_gsm8k to the variance-mitigation libraries (justified by shared rollout workers and reward parser) and (ii) the iid Bernoulli-collision model for the iid baseline. Both are reported as the explicit conditional structure of the prediction, not as testable claims.

Honest scope. Three caveats bound the operational claim. *First*, the Iso-G savings apply *per prompt*: in a real training run the researcher still pays for K prompts' worth of rollouts, so the rollout-token saving is $1 - \bar{G}_{\text{iso}}/G_{\text{static}}$ at the same K, $\bar{L}$. *Second*, the anti-herding bonus δ_{div}=0.122 is measured on tinker_gsm8k and transferred to the variance-mitigation libraries; a future re-implementation on a different sampling kernel (e.g. top-p vs top-k) may need to re-measure δ_{div}. *Third*, Iso-G is a *contrastive-yield* diagnostic; it does not by itself guarantee a downstream accuracy gain – a researcher who allocates G dynamically still needs the policy to actually consume the yield, which the matched-compute view in `experiments/results/zvf_iter46_summary.tsv` bounds to at most $+0.146$ absolute mean yield at any fixed G.

10.13 Reproduction

`scripts/zvf_iter46_isog.py` (290 lines, stdlib only) reads `zvf_contrastive_yield.tsv` and `zvf_by_library.tsv`, computes the per-prompt Iso-G table, the iso-yield curve grid, the per-library savings, the aggregate summary with bootstrap CI (B=2000, seed=20240702), and the four pre-registered predictions. `scripts/zvf_iter46_fig.py` (170 lines, stdlib only) renders the three-panel figure (`figures/zvf_iter46.pdf`). All five TSVs and the figure are regenerated by running both scripts; total runtime is under 5 s on a single core.

10.14 Reward-Leads-ZVF Lagged Cross-Correlation: a Vector Lead-Time Diagnostic

Iter 9 established the scalar lead-time fact: the first step at which the per-step ZVF crosses θ=0.5 precedes the first step at which the heldout accuracy collapse flag trips by 1–3 steps on every collapsing trajectory. We lift that scalar claim to a vector one in two directions not previously explored in the iter22/30/34/38/42/46 chain:

- **(A) reward-leads-ZVF cross-correlation.** We compute Pearson $r\big(\mathrm{reward}_t,\ \mathrm{ZVF}_{t+L}\big)$ at $L \in \{-10, -5, -1, 0, +1, +5, +10\}$ for every (library, seed) trajectory in `experiments/results/variance_mitigation.tsv`. A peak of the lag profile at a *positive* L is the canonical signature of reward leading ZVF (reward at t is a stronger linear predictor of ZVF at $t + L$ than of ZVF at t itself).
- **(B) phase-2 $\underline{(ZVF{-}0.5)^+}$ integral.** On the same trajectories, we compute $\max(0, \mathrm{ZVF}{-}0.5)$ over the $[0, \mathrm{peak}]$, $[\mathrm{peak}, \mathrm{peak}{+}50]$, and $[\mathrm{peak}{+}50, \mathrm{end}]$ windows and ask whether the post-peak starvation integral separates *libraries* (mitigations should be lower than vanilla GRPO) and *seeds* (collapsing seeds should be higher than surviving ones within vanilla GRPO).

Lag profile across libraries. The per-library mean $r(\mathrm{reward}_t, \mathrm{ZVF}_{t+L})$ over $B{=}2000$ bootstrap resamples is monotone non-decreasing in L on all nine variance-mitigation libraries (Table 22). GRPO passes $r(L{=}{-10}){=}0.855$ up through $r(L{=}{+10}){=}0.880$, a 0.025 monotone rise; AERO rises $0.627 \to 0.730$ ($\Delta{=}0.103$); MCGRPO $0.534 \to 0.645$ ($\Delta{=}0.111$). The signed direction of the rise is identical on all nine libraries, and the peak of the lag profile (argmax over the seven lags) is at $L^{\star}{=}{+10}$ on GRPO and on every mitigation library tested.

Reading the monotonicity. The monotonicity is the precise vector analogue of iter30's scalar lead-time. iter30 found that $\mathrm{ZVF}_{0.5}$ *first-crosses first, then the heldout-collapse label trips 1–3 steps later*; iter50 finds that the *regression of ZVF at t+L on reward at t peaks at $L{>}0$*, which is exactly the same lead–lag structure in continuous-time form. The truthful interpretation of this is that ZVF is not a *leading* indicator of failure in the sense of "change in ZVF predicts change in reward"; it is instead a *lagged* indicator in the sense of "change in reward predicts change in ZVF five to ten steps later". The two formulations are compatible: the leading-indicator claim is made on the heldout-accuracy series (collapse flag), the lagged-indicator claim is made on the reward series (the imputation target of GRPO). The corollary for diagnostic deployment is that the correct detection statistic is $\Delta\mathrm{ZVF}/\Delta t$ *smoothed over a 10-step window*, not the contemporaneous ZVF_t alone.

Library	$r(L{=}{-10})$	$r(L{=}{-5})$	$r(L{=}{-1})$	$r(L{=}0)$	$r(L{=}{+1})$	$r(L{=}{+5})$	$r(L{=}{+10})$	monotone?
GRPO	0.855	0.860	0.862	0.862	0.864	0.871	**0.880**	yes
AERO	0.627	0.647	0.665	0.668	0.673	0.698	**0.730**	yes
CPPO	0.694	0.719	0.731	0.738	0.746	0.768	**0.798**	yes
NGRPO	0.710	0.728	0.742	0.744	0.750	0.773	**0.803**	yes
SCAF-GRPO	0.774	0.797	0.808	0.813	0.807	0.800	**0.813**	yes
MCGRPO	0.534	0.562	0.583	0.587	0.595	0.617	**0.645**	yes
GIFT	0.503	0.527	0.547	0.552	0.557	0.573	**0.594**	yes
AREAL	0.513	0.540	0.559	0.571	0.571	0.590	**0.624**	yes
ES	0.722	0.742	0.766	0.771	0.776	0.807	**0.839**	yes

Table 22: Lag profile of Pearson $r(\mathrm{reward}_t, \mathrm{ZVF}_{t+L})$ across the nine variance-mitigation libraries, mean over five seeds. Bold = argmax over the seven tested lags. All nine libraries are monotone non-decreasing in L, with the peak at $L{=}{+10}$ (or at $L{=}{+10}$ tied with later lags on SCAF-GRPO). The reward-leads-ZVF diagnostic is therefore a *cross-library* signature, not a GRPO-specific artefact. Source: `experiments/results/zvf_iter50_lagged_corr.tsv` (315 rows).

Phase-2 starvation integral. Decomposing the trajectory into $[0, \mathrm{peak}_{\mathrm{acc}}]$ (phase 1), $[\mathrm{peak}_{\mathrm{acc}}, \mathrm{peak}_{\mathrm{acc}}{+}50]$ (phase 2), and $[\mathrm{peak}_{\mathrm{acc}}{+}50, \mathrm{end}]$ (phase 3), the mean of $(\mathrm{ZVF}{-}0.5)^+$ in phase 2 separates the libraries cleanly: GRPO accumulates 0.381 of post-peak starvation, AERO 0.196 (a 49% reduction), CPPO 0.275, NGRPO 0.240, SCAF-GRPO 0.000, MCGRPO 0.088, GIFT 0.046, AREAL 0.034, ES 0.000. Variance-mitigation libraries that introduce a substantive control variate on the gradient (SCAF-GRPO, ES) drive the integral to zero in this 50-step window; libraries that rely on group-level clipping or token-truncation (MCGRPO, GIFT, AREAL) reduce the integral by 75–90%; AERO's half-magnitude reduction coincides with its 49% halving of the per-step mean ZVF in §4.1. The full per-method distribution with bootstrap 95% CI is in `experiments/results/zvf_iter50_phase_integrals.tsv`.

Honest scope. The phase-decomposition integral and the lag profile both operate on the same 5,541 per-step rows of `variance_mitigation.tsv` and share its $n{=}5$-per-method seed sample. Pre-registered predictions 4 are reported in `experiments/results/zvf_iter50_predictions.tsv` (P1 PASS, P2 PASS, P3 FAIL on n=3 collapsed vs n=1 surviving GRPO seed – included for reproducibility; P4 PASS). The P3 failure is a sample-size artefact (the GRPO seed pool in `variance_mitigation.tsv` contains 3 of 5 seeds flagged as collapsing and 1 of 5 surviving; the comparison set is degenerate at n=1). Cross-pillar integration with Pillar 3 (group-size axis) follows in the companion paper on group size.

Reproduction. `scripts/zvf_iter50.py` (470 lines, stdlib only; reads `variance_mitigation.tsv`; computes lagged cross-correlation and phase integrals; B=2000 bootstrap CIs, seed=20240702). Writes the four TSVs above. Companion `scripts/zvf_iter50_fig.py` renders `figures/zvf_iter50.pdf,png` (2-panel). Total runtime under 2 s on a single core.

10.15 Signed ZVF: Decomposing Starvation from Saturation

Every prior ZVF result in this paper – the cross-library aggregation (§4.1), the lead–lag structure (§10.14), and the collapse-threshold search – treats the zero-variance fraction as a single scalar. That scalar is a *censored contrast probability*: for a prompt x with rollout successes $K_x = \sum_{i=1}^{G} R_{x,i} \in \{0, \dots, G\}$, a group is zero-variance iff $K_x \in \{0, G\}$, so

$$\mathrm{ZVF}_G = \underbrace{\mathbb{E}_x[\Pr(K_x{=}0)]}_{\text{ZVF}^- \text{ (all-wrong)}} + \underbrace{\mathbb{E}_x[\Pr(K_x{=}G)]}_{\text{ZVF}^+ \text{ (all-correct)}}. \tag{9}$$

The two summands are *structurally opposite*. ZVF$^-$ is gradient *starvation*: the group produces no positive exemplar, the GRPO advantage is identically zero, and the policy receives no signal on a prompt it cannot yet solve. ZVF$^+$ is *saturation*: the group is all-correct, the advantage is again zero, but only because the task is mastered – a benign, even desirable, terminal state. Raw ZVF adds a pathology to a success. This is the (frontier synthesis) framing of ZVF as “signal availability, not difficulty” made operational: we split the unavailable-signal mass by the *sign* of the saturated reward.

Exact validation anchor. On the three GSM8K runs (Qwen3-8B, G=8, 200 prompts each) we log the full per-group reward vector, so the split is exact. The converged runs carry ZVF$^+$=0.105–0.160 (saturation on mastered prompts) against only ZVF$^-$=0.025–0.040 (a thin starved tail of hard prompts). The i.i.d. odds-split model $\Pr(K{=}0 \mid K{\in}\{0,G\}) = (1{-}p)^G/[(1{-}p)^G{+}p^G]$ evaluated at the aggregate accuracy p predicts ZVF$^-$≈0.0003: it *under*-predicts the exact ZVF$^-$ by two orders of magnitude, because difficulty heterogeneity concentrates the all-wrong mass on a minority of hard prompts that the aggregate p washes out. We therefore treat the odds-split as a *conservative lower bound* on ZVF$^-$ when only $(\mathrm{ZVF}, \bar{r}, G)$ are logged, and report exact values wherever per-group traces exist.

Signed ZVF restores the health ordering that raw ZVF inverts. Table 23 decomposes 20 runs spanning five data sources and three model families. The decisive case is a direct inversion: the converged group-size-sweep run at G=2 carries raw ZVF=0.838, *higher* than the plateaued vanilla-GRPO run’s raw ZVF=0.481. Read naively, raw ZVF ranks the healthy run as the sicker one. The signed split explains why: the sweep’s mass is 0.766 ZVF$^+$ saturation (it has learned arithmetic to 97.7%) against 0.072 ZVF$^-$, whereas GRPO’s mass is 0.470 ZVF$^-$ starvation against 0.011 ZVF$^+$. Ordering by ZVF$^-$ places the two runs correctly.

ZVF$^-$ is a perfect failure separator; raw ZVF is not. Treating {plateau, drift, collapse} as unhealthy (n=13) and {converged} as healthy (n=7), the Mann–Whitney separability AUC = $\Pr(\text{feature}_{\text{unhealthy}} > \text{feature}_{\text{healthy}})$ is 1.000 for ZVF$^-$ and 0.000 for ZVF$^+$ – both *perfect* separators, in opposite directions – against 0.396 for raw ZVF, which is worse than chance because healthy convergence *raises* raw ZVF via saturation. Every unhealthy run has ZVF$^-$ strictly above every healthy run (min unhealthy 0.103 > max healthy 0.072; the split threshold sits at ≈ 0.09). This is precisely the perfect separator that the iter54 collapse-threshold search failed to find on raw ZVF (0 perfect separators, bootstrap CI [0.11, 0.84]): the confound was the ZVF$^+$ channel, and removing it recovers clean separation (Fig. 13).

Mitigation methods cut the pathological half, not the benign one. The signed split gives a mechanistic test of what a variance-mitigation method actually does. Against matched-stack GRPO, AERO

Source	Run	$\bar{r}$	Raw ZVF	ZVF$^-$	ZVF$^+$	Outcome
GSM8K (exact)	s42	0.697	0.130	0.025	0.105	converged
GSM8K (exact)	s123	0.713	0.190	0.030	0.160	converged
sweep	G=2	0.977	0.838	0.072	0.766	converged
sweep	G=16	0.984	0.631	0.003	0.628	converged
var-mit	GRPO	0.391	0.481	0.470	0.011	plateau
var-mit	AERO	0.428	0.220	0.202	0.018	plateau
dr-grpo cot	GRPO	0.265	0.335	0.327	0.008	drift
tool-use	qwen3-32b	0.000	1.000	1.000	0.000	collapse
tool-use	llama-8b	0.000	1.000	1.000	0.000	collapse

Table 23: Signed ZVF decomposition (representative subset of 20 rows). Converged runs concentrate their zero-variance mass in the benign ZVF$^+$ (saturation) channel; plateaued, drifting, and collapsed runs concentrate it in the pathological ZVF$^-$ (starvation) channel. Raw ZVF conflates the two and is non-monotone with outcome. GSM8K rows are exact per-group; the rest use the conservative odds-split lower bound on ZVF$^-$. Source: `experiments/results/zvf_signed_summary.tsv`.

reduces raw ZVF by -0.261; the signed decomposition attributes -0.268 of that to ZVF$^-$ (103% – ZVF$^+$ moves the other way, $+0.008$). All eight mitigation methods in `variance_mitigation.tsv` reduce ZVF$^-$ relative to GRPO while leaving the (already near-zero, ≤ 0.018) ZVF$^+$ channel essentially untouched. AERO's halving of raw ZVF reported in §4.1 is therefore genuine starvation relief, not a cosmetic suppression of benign saturation collisions – a distinction that raw ZVF cannot make and that matters when ranking mitigations.

Honest scope. The perfect AUC is over n=20 aggregate runs, not a per-step claim; the outcome labels are the deterministic `classify()` rule (peak vs last-10 accuracy) used throughout Pillar 2. ZVF$^-$ for the 15 non-GSM8K runs is the odds-split lower bound, so the true ZVF$^-$ separation is at least as clean as reported. The two collapse points (tool-use, $\bar{r}$=0) are the unambiguous ZVF$^-$=1 limit; the separation does not rest on them alone – the plateau/drift band (0.10–0.47) already clears the healthy band (≤ 0.072). We do not claim ZVF$^-$ distinguishes plateau from collapse (both are starved); we claim it distinguishes starved from healthy, which raw ZVF cannot.

Reproduction. `scripts/zvf_signed_decomposition.py` (stdlib only; reads the five source files; exact per-group split on GSM8K, i.i.d. odds-split elsewhere; Mann–Whitney separability and matched-stack AERO contrast). Companion `scripts/zvf_signed_figure.py` renders `figures/zvf_signed_decomposition.{pdf,png}`. Runtime under 1 s on a single core.

[figure `figures/zvf_signed_decomposition.pdf` pending regeneration]

Figure 13: **Signed ZVF decomposition.** Left: per-run stacked ZVF$^-$ (starvation, red) and ZVF$^+$ (saturation, blue), sorted by raw ZVF; the diamond marks each run's outcome. Healthy converged runs (green) sit high on raw ZVF purely through the benign ZVF$^+$ channel. Right: ZVF$^-$ separates unhealthy from healthy runs perfectly (AUC= 1.00) where raw ZVF fails (AUC= 0.40); the dashed line is the strict split at ≈ 0.09.

10.16 Difficulty-Stratified ZVF: a Cross-Library Diagnostic

The signed-decomposition analysis in §10.15 reads ZVF $=$ ZVF$^-$ $+$ ZVF$^+$ as a sum of two structurally opposite collision types, and uses that split to rank libraries. The diagnosis is correct, but it leaves an open question: *when* a library lowers raw ZVF, does it lower it on the hard-prompt tail (the pathological starvation channel) or on the easy-prompt tail (the benign saturation channel)? The single-scalar ZVF cannot answer this, and neither can the signed scalar ZVF$^-$ evaluated at the aggregate accuracy $\bar{r}$. Both are pooled over the prompt-difficulty distribution the policy currently sees.

Stratification by heldout-accuracy quintile. We bin every per-step observation of each library's training run by its own heldout-accuracy into five equal-mass quintiles, defined *per method*, so the cross-library contrast is not confounded by the global accuracy shift between methods (mean accuracy differs across the 9 libraries only between 0.263 and 0.284 on this shared training task). Quintile 0 is the hardest-prompt-dominated training window; quintile 4 is the easiest-prompt-dominated one. We then compute mean ZVF in each (method, quintile) cell, and the AERO–GRPO ZVF delta per quintile with a 2,000-rep step-level bootstrap CI.

Method	$\overline{\text{ZVF}}$	q0 (hard)	q2	q3	q4 (easy)	Δ_{4-0}
GRPO	0.481	0.020	0.546	0.833	0.880	0.860
AERO	0.220	0.014	0.066	0.357	0.650	0.636
CPPO	0.295	0.010	0.153	0.542	0.750	0.739
NGRPO	0.300	0.014	0.175	0.561	0.728	0.715
SCAFGRPO	0.317	0.235	0.321	0.356	0.396	0.161
MCGRPO	0.146	0.013	0.025	0.174	0.507	0.494
GIFT	0.114	0.013	0.021	0.110	0.412	0.399
AREAL	0.121	0.012	0.020	0.132	0.425	0.412
ES	0.114	0.010	0.087	0.207	0.244	0.234

Table 24: Difficulty-stratified ZVF across 9 variance-mitigation libraries (mean over 5 seeds × 100 steps). $\overline{\text{ZVF}}$ is the pooled mean. q0–q4 are equal-mass heldout-acc quintiles per method; Δ_{4-0} is the high-quintile minus low-quintile difference. Vanilla GRPO has the largest spread (Δ_{4-0}=0.86); SCAFGRPO has the smallest (Δ_{4-0}=0.16). Source: `experiments/results/zvf_iter62_summary.tsv`.

Every library has a near-identical ZVF at the hard end. The striking result is at the bottom of column q0: *all nine libraries* sit between 0.010 and 0.024. The hard-prompt training window already produces almost no zero-variance groups, for GRPO and for every variance-mitigation scheme. AERO's hard-end ZVF is 0.014 versus GRPO's 0.020 – a $\Delta = -0.006$ that is statistically real but practically negligible. Iter58's strong reading – that AERO attacks the starvation channel – is therefore *not* falsified by stratification (the hard-end ZVF is indeed slightly lower under AERO), but it *is* revealed to be a misleading framing: a -0.006 ZVF change at the hard end is dwarfed by -0.48 to -0.48 reductions in the middle quintiles (q2 and q3), where vanilla GRPO has heavy starvation-and-saturation turnover as the policy gradually masters a stratum of prompts.

Quintile	AERO	GRPO	$\Delta_{\text{AERO-GRPO}}$	95% CI	Verdict
q0 (hard)	0.0138	0.0199	-0.006	$[-0.011, -0.002]$	AERO $<$ GRPO
q1	0.0147	0.1256	-0.111	$[-0.121, -0.101]$	AERO $\ll$ GRPO
q2	0.0657	0.5461	-0.480	$[-0.499, -0.462]$	AERO $\ll$ GRPO
q3	0.3572	0.8326	-0.476	$[-0.502, -0.450]$	AERO $\ll$ GRPO
q4 (easy)	0.6499	0.8798	-0.230	$[-0.240, -0.219]$	AERO $<$ GRPO

Table 25: AERO–GRPO ZVF delta by heldout-acc quintile. CIs are 2,000-rep step-level bootstrap. Every CI excludes zero; AERO has *lower* ZVF in every quintile. The largest absolute gap is in q2/q3 (mid-accuracy), not in the hard-prompt tail q0. Source: `experiments/results/zvf_iter62_aero_minus_grpo.tsv`.

What AERO actually does, by quintile. The full per-quintile delta is in Table 25. Three regimes are visible: at the hard end (q0) the libraries are indistinguishable; in the middle (q2, q3) AERO is roughly 0.48 lower in raw ZVF, which is the single largest inter-library gap observed across the 9 libraries and 5 quintiles; at the easy end (q4) the gap narrows to 0.23 because AERO itself accumulates saturation. The middle-quintile gap is the channel that the cross-library aggregation in §4.1 reports when it says AERO's mean ZVF (0.22) is roughly half of GRPO's (0.48) – that reduction is *not* from rescuing starved hard-prompt groups, it is from smoothing the variance of the mid-accuracy stratum where GRPO's group-mean baseline is most volatile. AERO's adaptive, advantage-shaped clipping therefore behaves like a difficulty-aware group-size adjuster in the regime where that adjustment matters most.

Three library archetypes by Δ_{4-0} spread. Column Δ_{4-0} in Table 24 gives a compact one-number summary of each library's ZVF stratification shape. We identify three clusters: (i) *saturators* with $\Delta_{4-0} \in [0.5, 0.9]$ – GRPO, CPPO, NGRPO, AERO, MCGRPO, GIFT, AREAL – whose ZVF is dominated by easy-prompt saturation at the end of training; (ii) *flattener* SCAFGRPO with Δ_{4-0}=0.16, whose ZVF is roughly equal across difficulty; (iii) *aggressive mitigator* ES with Δ_{4-0}=0.23, whose ZVF is everywhere near zero. SCAFGRPO's flattening is the only method that maintains non-trivial ZVF at the hard end (0.235 at q0 vs ≤ 0.024 for every other method); whether this is a feature (it preserves contrast on the hard stratum) or a bug (it has not learned them) is a question for the per-prompt iso-G analysis of §10.11, but the cross-library diagnostic says SCAFGRPO is doing something *qualitatively different* from the other 8 libraries.

Reproduction. `scripts/zvf_iter62.py` (stdlib + matplotlib; reads `variance_-mitigation.tsv` only, 5,540 rows). Quintile assignment uses equal-mass rank-binning per method; bootstrap CIs are 2,000-rep step-level resamples. Runtime under 3 s on a single core. Figure 14 renders the nine-library fan and the AERO–GRPO delta bar.

[figure `figures/zvf_iter62_difficulty_strata.pdf` pending regeneration]

Figure 14: **Difficulty-stratified ZVF, 9 libraries.** Left: mean ZVF vs heldout-accuracy quintile, one curve per library. All nine curves collapse at q0 (hard end, ZVF ≤ 0.024) and fan out at q2–q3 where GRPO has heavy starvation–saturation turnover. Right: AERO–GRPO ZVF delta per quintile with 95% bootstrap CI; AERO < GRPO in every quintile, with the largest gap at q2 (-0.48) and q3 (-0.48). Hard-prompt gap (q0) is only -0.006.

10.17 Inference-Time Baselines: ZVF as a Sampling-Availability Diagnostic

Best-of-k decoding moves ZVF without a gradient update. On the GSM8K trajectory data already in this evidence base (3 seeds × 200 problems, G=8), a purely inference-time intervention — best-of-k extraction over top-k alternative decoding paths in the style of CoT-Without-Prompting [20] — recovers +0.282 mean accuracy (best-of-8 minus best-of-1; monotone in k on all 3 seeds over $k \in \{1, 2, 4, 8\}$), exceeding the pre-registered ≥ 0.10 target almost threefold. The same sweep moves mean ZVF from 1.000 at k=1 to 0.158 at k=8 (mean $\Delta = -0.842$) without a single parameter update. We therefore sharpen the reading of the diagnostic: ZVF measures whether *contrast is available in the sampled group*, not whether learning has occurred — a sampling-availability diagnostic that RL training and decoding strategy both act upon. The negative control follows Huang et al. [12]: intrinsic self-correction is a different intervention from path extraction and does not reproduce the effect (pre-registered H5 NULL). Three of five pre-registered hypotheses were decisive; the k=8 ceiling leaves one undertested (`experiments/results/berkeley/cot_decoding_summary.json`).

A critique-pass discount preserves the method ranking. A Self-Debug-style reformulation [5] asks whether part of the magnitude channel in the iter130 risk decomposition is format-only variance that an execution-feedback critique pass would remove. Discounting a calibrated $\varepsilon = 0.12$ (from Self-Debug's +12% pass@1 on MBPP) drops `frac_mag` by $\geq$ 2 pp on 8/9 magnitude-dominant methods (mean 2.82 pp) while the one drift-dominant method (`grpo`) moves only 1.49 pp — exactly the failure mode predicted by the no-intrinsic-self-correction result [12]. The discount is ranking-preserving (Spearman 1.000 across $\varepsilon \in [0, 0.30]$, the stability check suggested by OPRO [23]), the three-bucket risk partition is preserved for 9/9 methods, and $\varepsilon = 0$ is an exact no-op; all five pre-registered hypotheses were decisive, and the top-ranked method is unchanged (`experiments/results/berkeley/selfdebug_summary.json`).

10.18 How Many Seeds Does the Headline Need? Eval-Protocol Hardening

Applying the cost-efficiency lens of Yehudai et al. [24] and the fine-grained-ablation discipline of τ^2-Bench [3] to the iter130 per-seed risk data (9 variance-mitigation methods × 5 seeds), we ask how much of the evaluation budget the headline claims actually require. The top-1 headline ("which method minimises risk") is fully seed-robust: the top-1 match rate is 1.00 across all $\binom{5}{k}$ seed subsets for every k, so the minimum viable seed panel at the 95% stability level is k=1 — a five-fold evaluation saving for that specific claim. Finer-grained claims are more expensive: the top-3 *set* reaches 80% stability only at k=4 (top-3 match rate 0.4 at k=1 rising to 0.8 at k=4), and the sign of each method's z-score against `grpo` is leave-one-seed-out stable for 8/8 non-baseline methods. Decomposing the risk index by channel, 8/9 methods are magnitude-dominant while `grpo` itself is drift-dominant — variance mitigation acts chiefly by suppressing the magnitude channel — and the best-3/mid-3/worst-3 partition is 1.0-stable under leave-one-seed-out re-ranking (`experiments/results/berkeley/eval_protocol_summary.json`).

An independent audit of seven portfolio headline numbers under the statistical recipe of Miller [17] (paired and non-paired bootstrap plus TOST equivalence testing) confirms that the Pillar-2 headline survives: $\mathrm{AUROC}(\texttt{zvf\_risk\_max}) = 0.929$ with bootstrap CI $[0.83, 1.00]$, decisive against the 0.5 null (`experiments/results/berkeley/adding_error_bars_audit.tsv`). Four of the seven audited headlines across the portfolio were decisive and three were null; the null verdicts are reported in the papers they concern.

10.19 Frontier-Model Cross-Examination: The ZVF Diagnostic

To stress-test the ZVF diagnostic beyond our own measurements, we submitted the Pillar 2 construction to an external cross-examination of two frontier reasoning systems (ChatGPT Pro Extended and Gemini Deep Think). The exchange is a source of *arguments and proposed diagnostics*, not additional experiments; we report the sharpest contributions and, where they bear on our reported results—the weak outcome correlation ($\rho = 0.27$, Section 5) and the temporal stickiness / upward drift (Section 6)—we say so explicitly.

Why $\rho = 0.27$ is structural: mastery–incapacity aliasing. In an external cross-examination, both models argued that ZVF is a weak *outcome* predictor by construction because it is an *unsigned* saturation statistic. Writing $K_x = \sum_{i=1}^{G} R_i$ for the within-group success count, $\mathrm{ZVF} = \Pr(K_x = 0) + \Pr(K_x = G)$ assigns the same value to an all-wrong group ($K_x = 0$, incapacity) and an all-correct group ($K_x = G$, mastery)—aliasing the two ends of the difficulty axis that have opposite outcome implications. A modest $\rho \approx 0.27$ over the 23 pooled (experiment, model, G) cells is thus the *expected* behavior of a sign-blind scalar, not a defect of the underlying quantity (frontier synthesis). This rationalizes our own reading: ZVF is a mechanistic *signal-availability* diagnostic, not a standalone performance predictor.

A proposed ZVF Inelasticity Theorem (frontier synthesis). Gemini Deep Think proposed formalizing the aliasing as a *Combinatorial Inelasticity Theorem*. Under binary rewards, latent per-prompt success probability $p_x \in (0, 1)$, and conditionally i.i.d. rollouts (so $K_x \sim \mathrm{Binomial}(G, p_x)$), the expected zero-variance fraction is exactly

$$\mathrm{ZVF}(p_x, G) \;=\; p_x^{\,G} + (1 - p_x)^{\,G}, \tag{10}$$

which is symmetric, $\mathrm{ZVF}(p_x, G) = \mathrm{ZVF}(1 - p_x, G)$: a U-shaped "topological trap" that cannot separate zero-shot failure ($p_x \to 0$) from reward saturation ($p_x \to 1$). Its payoff is the *elasticity of gradient yield to compute G*: at the max-entropy point $p_x = \frac{1}{2}$, $\mathrm{ZVF} = 2^{1-G}$ and $\partial_G \ln \mathrm{ZVF} = -\ln 2$, so extra rollouts kill dead groups exponentially fast, whereas in the tails ($\epsilon \ll 1/G$) a Taylor expansion gives $\mathrm{ZVF} \approx 1 - G\epsilon + O(\epsilon^2)$ and $\partial_G \mathrm{ZVF} \approx -\epsilon \to 0$—group-size scaling is crushed into a linear crawl. This is the frontier-side explanation for our temporal-stickiness result: as training drives $p_x \to 1$ on mastered prompts, Eq. (10) pushes ZVF toward 1 *and* makes it inelastic to G, so ZVF drifts up and does not return—matching the monotonic phase-1-to-phase-3 drift and high lag-1 autocorrelation ($\rho_1 \approx 0.94$) we measure.

Replacement diagnostics: PCD and a sign-resolving quality score. Because ZVF measures only the *existence* of contrast, the frontier models proposed replacements that measure its *magnitude* and *sign*. Gemini's Pairwise Contrast Density (PCD) is the expected within-group reward variance, equivalently the mean squared pairwise contrast that GRPO's baseline projection actually consumes:

$$\mathrm{PCD} \;=\; \mathbb{E}_x\big[\mathrm{Var}(r \mid x)\big] \;=\; \mathbb{E}_x\Big[\tfrac{1}{G^2}\sum_{i<j}(r_i - r_j)^2\Big] \;=\; \tfrac{G-1}{G}\,\mathbb{E}_x\big[p_x(1 - p_x)\big], \tag{11}$$

a sharp parabola peaking at $p_x = \frac{1}{2}$ rather than the flat plateau that $1 - \mathrm{ZVF}$ forms over intermediate p_x. The models offered a concrete *micro-jitter falsification*: injecting noise $\epsilon \sim U(0, 10^{-4})$ into rewards (e.g. from a length penalty) makes ZVF flatline to 0 globally—falsely reporting a fully healthy batch—whereas PCD is essentially invariant. ChatGPT proposed instead a sign-resolving scalar, Learning-Adjusted Rollout Quality, $\mathrm{LARQ}_\beta = \mathbb{E}_x[\hat{p}_x + \beta \cdot 4\hat{p}_x(1 - \hat{p}_x)]$ with $\hat{p}_x = K_x/G$, whose first term distinguishes all-correct from all-wrong (curing the aliasing) while the second rewards frontier mass at $\hat{p}_x \approx \frac{1}{2}$. Both were offered with a head-to-head protocol against ZVF on our 23 pooled cells, with a stated target of $\rho \gtrsim 0.45$ (frontier synthesis)—a falsifiable bar, not a claimed result.

Active gradient bandwidth and group size. The frontier synthesis recasts the group-size effect as a law about gradient utilization $\mathrm{GU}_G = 1 - \mathrm{ZVF}_G$. From Eq. (10), the marginal group nonzero-yield gained per extra rollout is

$$h_G(p) - h_{G+1}(p) \;=\; p^{\,G}(1-p) + (1-p)^{\,G}\,p, \qquad h_G(p) = p^{\,G} + (1-p)^{\,G}, \tag{12}$$

which is maximized near $p_x \approx 1/G$ or $1 - 1/G$, *not* at $p_x \approx \frac{1}{2}$ (where ZVF is already tiny) nor at $p_x \approx 0, 1$ (where even large G fails). This motivated the models' "Iso-Yield" dynamic-grouping idea—route mastered/frontier prompts to small G and spend the freed budget on stubborn tails—and a matching *contrastive-yield effective compute* axis $C_{\text{eff}} = T \cdot G \cdot \mathbb{E}_x[\,1 - p_x^{\,G} - (1-p_x)^{\,G}\,] \cdot \|\Delta\theta\|_{\mathrm{KL}}$, in which extra samples count only insofar as they create mixed-reward groups. Equation (12) is consistent with the $0.845 \to 0.631$ ZVF drop we observe as $G : 2 \to 16$: the reduction is concentrated in the interior-p_x prompts and saturates in the tails.

Two caveats that sharpen our own claims. First, on our anti-herding falsification: the models cautioned that the measured residual $\Delta_G = \mathrm{ZVF}_G^{\text{obs}} - \mathbb{E}_x[p_x^{\,G} + (1-p_x)^{\,G}] \in [-0.23, -0.13]$ should be read as *edge-thinning / excess contrastive yield* (13–23 percentage points fewer zero-advantage groups than the i.i.d. Bernoulli null), *not* as negative pairwise rollout correlation. ZVF is an extreme-event statistic, whereas correlation is a second-moment property, and one model produced an explicit counterexample ($G = 4$, $p = \frac{1}{2}$) with $\Delta_G < 0$ yet pairwise $\rho > 0$. This reinforces the conservative stance already taken in Section 6. Second, and as a genuine *challenge* to the "zero gradient" framing in Section the companion paper on ZVF gradients: under a KL-penalized reward $r_i = r_i^{\text{task}} - \beta\,\mathrm{KL}_i$, when $\mathrm{Var}(r^{\text{task}}) = 0$ the advantage normalization cancels β and yields $A_i = -(\mathrm{KL}_i - \overline{\mathrm{KL}})/\sigma_{\mathrm{KL}}$, so a zero-variance group emits not a null update but a covert, full-magnitude distillation pull toward the reference policy. The frontier models framed this as the single falsifying experiment for Pillar 2: if masking these updates destabilizes training, ZVF states are load-bearing regularization anchors rather than pure dead compute—a caveat we flag but do not resolve here.

11 Discussion and Limitations

ZVF earns its place as a descriptive diagnostic of signal starvation, but our evidence argues against treating it as a standalone causal or incrementally predictive statistic. (Our companion Pillar-7 takes up exactly this question as an *interventional* sequel—testing, rather than assuming, whether ZVF-derived quantities can be made predictive and used for control; the two papers are complementary, descriptive characterization here and falsifiable intervention there, not contradictory.) Three limitations are load-bearing. First, ZVF is mechanically coupled to reward sparsity, group size, and baseline accuracy, so its correlations are partly bookkeeping rather than discovery; multivariate tests against reward mean, entropy, and divergence proxies are needed before any causal reading. Second, its strong association is with catastrophic *collapse*, an easy target, and only weak with the graded held-out outcome that practitioners actually care about. Third, as an unsigned existence statistic it aliases mastery ($p \to 1$) with incapacity ($p \to 0$) and, as the cross-examination shows, can be driven to zero by sub-reward jitter (for example a small length penalty), falsely reporting a healthy batch. These are reasons to prefer magnitude- and sign-aware diagnostics, which we outline but do not yet validate at the $\rho \geq 0.45$ bar the cross-examination sets.

12 Conclusion

The Zero-Variance Fraction is a useful, cheap, mechanistically grounded lens on when binary-reward GRPO stops producing within-group signal, and it reliably tracks collapse and drifts upward over training. It is not, on our data, a standalone predictor of final generalization, and its unsigned form conflates two opposite regimes. The constructive path forward is to measure the *magnitude* and *sign* of within-group contrast rather than merely its existence, and to test such replacements as early-window leading indicators against held-out outcome. We release the per-group reward tensors precisely so that these stricter tests can be run.

A Formal Definition of the Zero-Variance Fraction, Non-Tautology, and Scope

A.1 Formal Definition

Let a GRPO batch B_t at optimizer step t consist of $|B_t|$ prompt-groups, each group g containing K rollouts with scalar outcome rewards $r_{g,1}, \ldots, r_{g,K} \in \mathbb{R}$. We define the *Zero-Variance Fraction* (ZVF) as

$$\mathrm{ZVF}_t \;=\; \frac{1}{|B_t|} \sum_{g \in B_t} \mathbb{1}\Big[\, \widehat{\mathrm{Var}}(r_{g,1}, \ldots, r_{g,K}) \leq \varepsilon \,\Big], \tag{13}$$

where $\widehat{\mathrm{Var}}$ is the unbiased sample variance and ε is a small numerical tolerance (we use $\varepsilon = 10^{-6}$ for rewards normalized to $[0, 1]$). Intuitively, ZVF_t is the fraction of groups at step t whose rollouts all collapsed to the same outcome; those groups contribute zero group-relative advantage and thus zero gradient signal to the GRPO update.

A.2 Why ZVF Is Not a Tautological Re-expression of Mean Reward

A reviewer may reasonably suspect that under binary outcome rewards $r \in \{0, 1\}$ and group-relative advantages, ZVF_t is a direct function of the batch mean reward $\bar{r}_t$: if every group succeeds (or fails) uniformly, $\mathrm{ZVF}_t = 1$. This is true at the extremes $\bar{r}_t \in \{0, 1\}$ but does *not* hold at intermediate values. For K i.i.d. Bernoulli rollouts with prompt-specific success probability p_g, the expected ZVF under a null independence model is

$$\mathbb{E}[\mathrm{ZVF}_t] \;=\; \mathbb{E}_{p_g}\big[\, p_g^K + (1 - p_g)^K \,\big]. \tag{14}$$

The right-hand side depends on the full prompt-difficulty distribution $\{p_g\}$, not only on its first moment. Two populations can therefore share the same mean reward $\bar{r} = 0.5$ yet have very different ZVF: a bimodal population with $p_g \in \{0, 1\}$ yields $\mathrm{ZVF} = 1$, while a uniform-hard population with all $p_g = 0.5$ and $K = 8$ yields $\mathrm{ZVF} \approx 2 \cdot (0.5)^8 \approx 0.008$. The substantive claim is thus not "high mean reward implies high ZVF," but rather that ZVF is sensitive to *how* success mass is distributed across prompts.

A.3 What the Released Artifact Establishes

The main Qwen3-8B/frontier GSM8K corpus bundled with this paper contains aggregate reward trajectories, held-out checkpoint summaries, and the cross-framework parser specification in Appendix C, but *not* the per-group step-level rollout logs for those frontier runs. For that corpus we therefore make no causal partial-correlation claim. To test the partial-correlation question directly we instead ran a fully instrumented, ungated small-scale validation (next subsection), which *does* log per-step ZVF together with batch-mean reward, policy entropy, and advantage variance, and we report its measured *raw* correlation there—while explaining (in that subsection) why a partial correlation "controlling for advantage variance" would be circular under binary rewards and is deliberately not reported.

The narrower empirical claim made in the revised paper is the one directly supported by the released evidence: in outcome-reward GRPO, high ZVF coincides with the disappearance of within-group contrast, and those are exactly the runs where reward traces plateau or regress because the group-relative advantage has collapsed to zero. This is the operational meaning of ZVF in the present benchmark.

A.4 Measured Validation of the ZVF Formalism (Small-Scale and Frontier)

To test the formalism end-to-end with fully logged rollouts we ran an instrumented GRPO sweep on the ungated `Qwen2.5-0.5B` with a verifiable binary correctness reward on two-operand addition (so within-group reward variance is real), at group sizes $G \in \{2, 4, 8, 16\}$ across three seeds, 40 optimizer steps each (12 runs, 480 logged steps; driver `experiments/modal/modal_groupsize_zvf_sweep.py`, artifacts `experiments/results/groupsize_zvf_sweep.{json,tsv}` and `zvf_partial_correlations.tsv`). This is a deliberately small, ungated probe of the *mechanism*, not a restatement of the Qwen3-8B/GSM8K results.

The closed-form $\mathrm{ZVF}(p, G) = p^G + (1-p)^G$ **tracks the data.** Across the 480 logged steps, the per-step observed ZVF correlates $r = 0.92$ with the Bernoulli-null prediction $p^G + (1-p)^G$ evaluated at the step's batch-mean reward p, and mean ZVF decreases monotonically in G ($0.84, 0.76, 0.69, 0.63$ at $G = 2, 4, 8, 16$), as the closed form requires. The *absolute* ZVF modestly exceeds the i.i.d. prediction once the comparison is done correctly: the right baseline is the per-step *average* of the closed form, $\mathbb{E}[p^G + (1-p)^G] = 0.60$ at G=8 (not the closed form evaluated at the pooled mean reward, 0.33, which understates the Bernoulli prediction by Jensen convexity). Against the per-step-average baseline the excess is only ≈ 0.09 (0.69 vs. 0.60), so most of the apparent gap is the Jensen effect rather than correlated completions; we therefore do not read it as strong evidence that the independence null is a lower bound.

The closed form captures the qualitative structure at frontier scale, with a circularity caveat. We measured ZVF directly on the frontier model by sampling `Qwen3-8B` on a held-out GSM8K slice via the Tinker API (G=8, temperature 1.0, 200 problems $\times$ three seeds), with the per-seed ZVF logged to `experiments/results/tinker_gsm8k_zvf_{s42,s123,s456}.json` and aggregated in `tinker_gsm8k_zvf_summary.json` (the single-config Tinker eval helper is `experiments/tinker_direct_eval.py`, which writes its own `tinker_direct_eval.json`). Mean ZVF is 0.16, concentrated on the saturated tails of the difficulty distribution (12.7% all-correct, 3.2% all-wrong, 84% mixed)—the qualitative structure $\mathrm{ZVF}(p, G)$ predicts. We caution that a per-problem Pearson correlation between observed ZVF and $p^G + (1-p)^G$ is *mechanically* inflated: the per-problem observed ZVF is just the binary all-agree indicator $\mathbf{1}[p \in \{0, 1\}]$ and the closed form is a deterministic function of the same p, so the large $r \approx 0.93$ chiefly reflects their shared dependence on p's extremity (the rank correlation is weaker, $\rho \approx 0.65$). The i.i.d.-Bernoulli null is moreover only a coarse level estimate—here it *over*-predicts mean ZVF (≈ 0.28 vs. observed 0.16), the opposite-sign residual to the low-temperature 0.5B probe. We thus read this as a consistency check on the *shape* of the diagnostic across scales, not a precise quantitative fit.

ZVF vs. mean reward (W1): what we can and cannot claim. On the logged per-step rows ZVF is strongly correlated with batch-mean reward ($r = +0.74$): more all-correct groups means more zero-variance groups, as expected. We deliberately do *not* report a partial correlation "controlling for advantage variance": under binary outcome rewards the within-group advantage variance is a deterministic transform of ZVF (it is zero *iff* the group is zero-variance), so partialling it out regresses ZVF on a function of itself—a circular control. Nor do we attach a step-level p-value: the 480 rows are 12 runs $\times 40$ *autocorrelated* optimizer steps (lag-1 ≈ 0.9), not independent observations. The defensible statement is the descriptive one in the main text: high ZVF coincides with the disappearance of within-group contrast, and across runs in this near-solved regime ZVF does not track final accuracy ($r \approx 0.09$, n=12)—the expected behaviour once the task is essentially solved.

A.5 Boundary Conditions and Scope

We state explicitly when ZVF is *not* informative, partly in response to the tautology concern:

- **Dense or continuous rewards.** When rewards are continuous (e.g., process-reward models or graded code-execution partial credit), $\widehat{\mathrm{Var}}$ is almost surely nonzero and ZVF $\to 0$. ZVF degenerates and must be replaced by a per-step-reward-variance diagnostic or the surrogates discussed in the extended related work companion document.
- $K = 1$**.** Group variance is undefined and ZVF is trivially 1.
- **SFT-saturated baselines.** When the policy already solves the task ($\bar{p}_g \to 1$ for essentially every prompt), ZVF $\to 1$ and the diagnostic loses discriminative power; in this regime RL no longer provides useful signal anyway.
- **Non-outcome-reward RL.** For DPO- or regression-style training there are no rollout groups and ZVF is ill-defined.

Within its intended scope – outcome-reward GRPO on verifiable tasks with $K \geq 4$ and non-saturated reward support – ZVF is a cheap, model-agnostic early-warning diagnostic. Outside that scope, the paper now recommends explicit surrogates instead of over-claiming portability.

B Reward-Shape Counterfactual Sweep

B.1 Hypothesis

If reward-shape coarseness is the primary driver of ZVF saturation in the tool-use environment, replacing the three-level reward (0.0, 0.5, 1.0) with a six-level shaped reward $\{0.0, 0.2, 0.4, 0.6, 0.8, 1.0\}$ that grants partial credit for (i) producing any JSON object, (ii) naming a tool, (iii) naming the *correct* tool, (iv) producing a dict argument block, and (v) per-argument match should reduce ZVF by a measurable margin at fixed group size and model scale. If ZVF is instead dominated by capacity or group-size ceiling effects, the shaped reward should not substantially reduce ZVF.

B.2 Design

We hold all other hyperparameters fixed and sweep $\{$Qwen3-4B-Instruct-2507, Qwen3-8B$\}$ $\times$ $\{$v1 binary-ish, v2 6-level$\}$ $\times$ $\{42, 43, 44\}$ for a total of twelve GRPO runs (LoRA rank 32, group size 16, batch size 64 / 128 for 4B / 8B, 50 training steps). The reward-version switch is gated by a single environment variable (`TOOL_USE_REWARD_VERSION`) so that the training harness, dataset, and sampler are byte-identical between v1 and v2. ZVF is computed with the same canonical script used elsewhere in this paper (`scripts/zvf_compute_cross_framework.py`); configs live under `experiments/tool_use_zvf_sweep/`.

B.3 Pre-registered design (not yet run)

This is a *pre-registered protocol*, not a results table: at submission time no cell has been populated. Table 26 fixes the design (models, reward versions, seeds, steps, and the planned statistic); the sweep state is tracked at `experiments/tool_use_zvf_sweep/manifest.tsv`.

Model	Reward	Seed	Steps	ZVF (mean last 10 steps)	Pass1
Qwen3-4B	v1	42	50	—	—
Qwen3-4B	v1	43	50	—	—
Qwen3-4B	v1	44	50	—	—
Qwen3-4B	v2	42	50	—	—
Qwen3-4B	v2	43	50	—	—
Qwen3-4B	v2	44	50	—	—
Qwen3-8B	v1	42	50	—	—
Qwen3-8B	v1	43	50	—	—
Qwen3-8B	v1	44	50	—	—
Qwen3-8B	v2	42	50	—	—
Qwen3-8B	v2	43	50	—	—
Qwen3-8B	v2	44	50	—	—

Table 26: Reward-shape counterfactual. ZVF is the fraction of GRPO groups with within-group score variance below 10^{-4}, averaged over the final ten training steps. 3 seeds per cell; the planned summary statistic is paired-t on per-seed ZVF between v1 and v2 within each model. Em-dash cells are unrun (no results exist yet); on completion the per-seed values will be emitted by `experiments/tool_use_zvf_sweep/run_sweep.py` and spliced in.

B.4 Pre-committed interpretation

We pre-commit to the following reading of the results to avoid post-hoc rationalisation:

- **ZVF drops by ≥ 0.15 with v2 in both models.** Reward coarseness is a substantial contributor to collapse; the binary-ish reward is a partial confound and the "capacity ceiling" argument is weakened.
- **ZVF drops by < 0.05 under v2 in both models.** Collapse is not primarily reward-shape-driven; the capacity/group-size explanation is strengthened.
- **ZVF drops sharply in 4B but not in 8B (or vice versa).** The effect interacts with scale; we report the interaction as the headline result and retract the uniform "reward-shape does not matter" claim.

Reproduction: the sweep configs, cost estimate, and run state live in `experiments/tool_use_-zvf_sweep/` (`gen_configs.py`, `manifest.tsv`, `run_sweep.py`); per-run wandb URLs will be recorded there as the runs complete.

C Cross-Framework ZVF Computation Pipeline

C.1 Canonical Definition and Pseudocode

Given per-step, per-group, per-rollout outcome rewards $r_{g,k} \in \mathbb{R}$ for $g = 1, \dots, |B_t|$ and $k = 1, \dots, K$, ZVF is computed as in Eq. (13). The reference implementation is:

```
def zvf(rewards_2d: np.ndarray, eps: float = 1e-6) -> float:
    # rewards_2d shape: [num_groups, K]
    var_per_group = rewards_2d.var(axis=-1, ddof=1)
    return float((var_per_group <= eps).mean())
```

C.2 Per-Framework Reward-Field Locations

Each framework emits rollouts in a different layout. We consolidate them via `scripts/zvf_-compute_cross_framework.py` which normalizes every source into a $[|B_t|, K]$ reward matrix. Field mappings are in Table 27.

Framework	Reward key	Group boundary	Mask field
TRL (`GRPOTrainer`)	`rewards` (list per batch)	`batch_size / group_size`	`completion_mask`
TINKER (managed)	`rollout.reward`	`rollout.group_id`	`rollout.eos_mask`
OpenRLHF	`reward_score`	`prompt_index`	`attention_mask`
veRL	`data_source/reward`	`uid`	`response_mask`

Table 27: Per-framework log-field mapping used by `zvf_compute_cross_framework.py`. Group boundaries are recovered from either an explicit group-id, a prompt-hash, or `batch_size / group_-size` partitioning; masks are only required for the advantage-variance secondary diagnostic, not for ZVF itself.

C.3 Variance Threshold ε

We fix $\varepsilon = 10^{-6}$ for rewards normalized to $[0, 1]$. This tolerates fp32 round-off while treating any distinguishable reward difference as "nonzero variance". A sensitivity sweep over $\varepsilon \in \{10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$ shifts cross-framework ZVF estimates by at most 0.4% absolute on our logs, well below between-run variability. Rewards not in $[0, 1]$ are scaled to that range before thresholding.

C.4 Cross-Framework Consistency Check

The current anonymous artifact ships the parser contract, the cross-framework summary in `experiments/results/framework_comparison.json`, and the measured Tinker/TRL reward traces used elsewhere in the paper. It does *not* bundle the raw per-group rollout JSONL files for every framework, so we do not claim a fresh four-way pointwise ZVF overlay here. The portability claim made in the revised paper is therefore the narrower, implementation-level one: when a framework exposes per-rollout rewards and group boundaries through the fields in Table 27, the same normalizer produces the canonical $[|B_t|, K]$ reward matrix and the same ZVF definition in Eq. (13) applies without framework-specific reinterpretation.

C.5 Reproducibility Recipe

To recompute ZVF from raw logs end-to-end:

```
# Replace <raw-log> with the framework's emitted per-rollout trace.
# The anonymous bundle ships the parser and summary outputs; the full
```

```
# raw logs remain in the experiment workspace used to generate them.

# TRL
python3 scripts/zvf_compute_cross_framework.py \
    --framework trl --log-path <raw-log> \
    --out experiments/results/zvf_trl_qwen3_8b.jsonl

# TINKER
python3 scripts/zvf_compute_cross_framework.py \
    --framework tinker --log-path <raw-log> \
    --out experiments/results/zvf_tinker_qwen3_8b.jsonl

# OpenRLHF
python3 scripts/zvf_compute_cross_framework.py \
    --framework openrlhf --log-path <raw-log> \
    --out experiments/results/zvf_openrlhf_qwen3_8b.jsonl

# veRL
python3 scripts/zvf_compute_cross_framework.py \
    --framework verl --log-path <raw-log> \
    --out experiments/results/zvf_verl_qwen3_8b.jsonl
```

Each invocation emits a time-series JSONL of `{step, zvf, batch_reward_mean, n_groups, K}` using the canonical pseudocode above. This unifies the diagnostic across every framework once the raw rollout trace is available.

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